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Threshold for neoclassical magnetic islands in a low collision frequency tokamak

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A kinetic theory for magnetic islands in a low collision frequency tokamak plasma is presented. Self-consistent equations for the islands' width, w , and propagation frequency, ω , are derived. These include contributions from the perturbed bootstrap current and the toroidally enhanced ion polarization drift. The bootstrap current is independent of the island propagation frequency and provides a drive for the island in tokamak plasmas when the pressure decreases with an increasing safety factor. The polarization drift is frequency dependent, and therefore its effect on the island stability cannot be deduced unless ω is known. This frequency is determined by the dominant dissipation mechanism, which for low effective collision frequency, $\nu_{\text{eff}} = \nu/\epsilon < \omega$, is governed by the electrons close to the trapped/passing boundary. The islands are found to propagate in the electron diamagnetic direction in which case the polarization drift is stabilizing and results in a threshold width for island growth, which is of the order of the ion banana width. At larger island widths the polarization current term becomes small and the island evolution is determined by the bootstrap current drive and Δ' alone, where Δ' is a measure of the magnetic free energy. [S1070-664X(96)02601-7]

I. INTRODUCTION

Single-fluid, resistive magnetohydrodynamics (MHD) predicts that magnetic islands will grow at a rate proportional to the tearing mode parameter, Δ' , which measures the free energy available for the resistive reconnection of magnetic field lines. It is evaluated from the marginal ideal MHD equations and depends on the current profile in the plasma.¹ Typically Δ' is negative for magnetic perturbations that have a poloidal mode number $m \geq 2$, so that such perturbations are damped. However, when the nonlinear effects associated with the presence of the magnetic island itself are taken into account, additional instability drives for the island can exist. Two particular mechanisms that could play a role have been investigated. First, the self-consistent deformation of density and temperature profiles associated with the magnetic island perturbs the bootstrap current, and this effect provides a drive for the island in tokamak plasmas when $dp/dq < 0$.^{2,3} Here p is the plasma pressure and q is the safety factor. Second, finite ion Larmor radius (FLR) effects can destabilize the island.⁴⁻⁹ In the limit that the island width is larger than the ion Larmor radius, the dominant FLR contribution to the perturbed current is the ion polarization current, which arises because of the time variation of the electric field associated with the propagating islands. The polarization current therefore depends on the propagation frequency, ω , and its effect on the island growth (i.e., whether it is stabilizing or destabilizing) cannot be determined until ω is known. Indeed, because the expression for ω is obtained from toroidal torque balance, its determination is rather subtle and depends on the details of the dissipation processes in the plasma and "external" forces, for example those resulting from error fields.

Early neoclassical theories of magnetic island evolution^{2,3} neglected the effects of ion inertia in deriving an expression for the island growth due to the bootstrap current perturbation. This drive was predicted to produce magnetic islands, even when resistive MHD predicts stability ($\Delta' < 0$). In the absence of any other effect, it was suggested that islands of all widths (down to the linear, resistive layer width) should exist in tokamaks and produce anomalous transport.¹⁰ While there is sometimes evidence for the existence of small-scale magnetic islands,^{11,12} bootstrap current-driven islands do not seem to be pervasive.

Recent studies of MHD activity in supershots on the Tokamak Fusion Test Reactor (TFTR) [Fusion Technol. **21**, 1324 (1992)] have yielded impressive agreement between the predictions of single-helicity neoclassical island formation theory and observations of $m/n = \frac{3}{2}, \frac{4}{3}, \frac{5}{4}$ islands, where Δ' is expected to be stabilizing.¹³ It was found that the bootstrap current drive theory accurately describes the temporal evolution and saturated island widths in TFTR, once the observed island exceeded a threshold width of order 1 cm. Thus, the neoclassical theory appears to be relevant to a description of magnetic island evolution, though extensions to the existing theories are needed to account for the threshold. The purpose of this paper is to address the threshold: two particular mechanisms for this have been proposed in the literature.

First, there is the possibility that the neoclassical polarization current could provide a stabilizing effect.^{14,15} For island widths that exceed the trapped ion banana width, this contributes a term to the island evolution equation that is inversely proportional to the cube of the island width, and will therefore be the dominant term for small widths (although these are assumed in the theory to be greater than the banana width). As noted above, the effect of this term on the island stability is dependent upon the island propagation frequency, ω . This frequency is not determined in Ref. 15, but,

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under the assumption that the term is stabilizing, the threshold is shown to have some of the features observed in the TFTR experiment. Furthermore, for larger islands the polarization current becomes small and the island evolution is then expected to be dominated by the balance between the bootstrap current drive and the stabilizing effect of Δ' .

A second mechanism for a threshold island width relates to the radial transport processes in the tokamak. For large island widths, w , above this threshold, the radial transport in the island region is exceeded by the parallel transport along the perturbed field lines, which causes density profiles to be flattened across the island. The resulting bootstrap current drive is inversely proportional to the island width and the island evolves to a stable saturated solution. However, as shown in Refs. 16 and 17, for a smaller island width the perpendicular transport dominates the parallel and the particle density is not completely flattened across the island. The island drive is then found to be proportional to w and there are no stable solutions for w . Thus, this theory also provides a threshold width for the existence of a stable solution for the island.

In this paper we concentrate on the role of the polarization current in the island evolution. Earlier works^{14,15} have analyzed the problem in a collisional fluid, relevant when the particle collision frequency for species j , ν_j , exceeds either the propagation frequency, ω , or the parallel streaming, $k_{\parallel}v_{\parallel j}$, whichever is the larger. Here $v_{\parallel}$ is the particle velocity parallel to the magnetic field and $k_{\parallel} = k_{\theta}w/L_s$, with $k_{\theta} = m/r$ the poloidal wave number and L_s the shear length ($L_s = R/q$, where R is the major radius, q is the safety factor, and s is the magnetic shear). The collisional fluid approximation is not generally valid in present-day tokamaks; certainly not for parameters typical of TFTR discharges. We therefore calculate the effect of the polarization current for a low collision frequency plasma, where $\nu_j < \epsilon\omega$. In such a plasma the dissipation is found to be dominated by those electrons within a narrow layer of pitch angle around the trapped/passing boundary. Neglecting the effects of sheared plasma flows, which are beyond the scope of this paper, and external torques, such as those from error fields, we find that toroidal torque balance yields a mode frequency $\omega = \omega_{*e}(1 + \eta_e/4)$, where ω_{*e} is the electron diamagnetic frequency. The polarization current is then found to be stabilizing and a threshold island width results from balancing this effect against the bootstrap current drive. The main result of this paper is that this threshold island width is

$$w_c = \epsilon^{1/2} \left(\frac{r}{sL_n} \right)^{1/2} \left(\frac{(1 + \eta_e/4)[\tau(1 + \eta_e/4) + 1 + \eta_i]}{1 + \eta_e/4 + \tau^{-1}(1 - \eta_i/2)} \right)^{1/2} \rho_{\theta i}, \quad (1)$$

where ϵ is the inverse aspect ratio, $L_n = -(d \ln n/dr)^{-1}$ is the density gradient scale length, $L_{Tj} = -(d \ln T_j/dr)^{-1}$ is the temperature scale length of the species j , $\eta_j = L_n/L_{Tj}$, $\tau = T_e/T_i$ is the ratio of electron to ion temperature, $\rho_{\theta i} = v_{thi}/\omega_{\theta i}$ is the ion poloidal Larmor radius with $\omega_{\theta i}$ the cyclotron frequency calculated using the poloidal component of the magnetic field only, and $v_{thj} = \sqrt{2T_j/m_j}$ is the thermal velocity of species j .

It is helpful to discuss our analytic approach in the collisionless regime and relate it to the fluid regime of earlier calculations.^{14,15} For the electrons we employ a drift-kinetic equation to describe their response to the magnetic and associated electrostatic perturbations. The island width, w , is assumed to be small compared to the minor radius of the rational surface, r , and large compared to the electron poloidal Larmor radius, $\rho_{\theta e}$. The electron kinetic equation is solved by a double expansion in the small parameters w/r and $\rho_{\theta e}/w$ in a manner similar to that of Ref. 2. However, we order the effective collision frequency, ν_e/ϵ , to be smaller than the propagation frequency, ω . The ion response is also calculated from the drift-kinetic equation. For this species we perform a double expansion in the small parameters w/r and $\epsilon^{1/2}\rho_{\theta i}/w$, where $\epsilon^{1/2}\rho_{\theta i}$ is the trapped ion banana orbit width. The condition that this second parameter be small places constraints on the validity of the threshold criterion derived above; we shall discuss this further in the Conclusion.

We can describe the nature of the solution for the particle distribution function qualitatively if we neglect the effects of equilibrium density and temperature gradients and consider the response to an electrostatic perturbation. This helps in understanding the origin of the neoclassical polarization current in the different collision frequency regimes. As noted in Ref. 18, in the presence of a radial electric field trapped particles will experience a nonzero bounce-averaged parallel flow $u = cE_r/B_{\theta}$ when averaged over a banana orbit. This ensures the poloidal projection of u cancels the poloidal component of the $\mathbf{E} \times \mathbf{B}$ drift velocity, so that there is no net drift of banana orbits in the poloidal direction. The result is a toroidal bounce-averaged precessional drift of the trapped particles, approximately equal to u . The trapped particles will therefore adopt a Maxwellian velocity distribution centered about $v_{\parallel} = u$, where we assume $u \ll \epsilon^{1/2}v_{th}$, so that the peak in the distribution lies within the trapped region of phase space. This effect is a consequence of the large radial excursion of the trapped particles during a bounce orbit, and it is therefore not important for the passing particles. To demonstrate this claim, we consider the change in kinetic energy of trapped particles, as they do work against the radial electric field in passing a radial distance equal to their banana width. Figure 1 illustrates a trapped particle executing a banana orbit of width ρ_b in the presence of a radial electric field, E_r . In traveling from point 1 to point 2, the work done is

$$W = q_j E_r \rho_b,$$

and there is a change in the particle kinetic energy equal to W . Assuming that the magnetic moment is conserved, and that the change in the magnetic field is negligible between points 1 and 2,

$$v_{\parallel} m_j (\Delta v_{\parallel}) = q_j E_r \frac{v_{\parallel}}{\omega_{\theta}},$$

where we have used $\rho_b = v_{\parallel}/\omega_{\theta}$. Here $\Delta v_{\parallel}$ is the change in the particle parallel velocity between points 1 and 2 and corresponds to the bounce-averaged flow. The above equation yields $\Delta v_{\parallel} = u$, as required. The passing particles do not drift

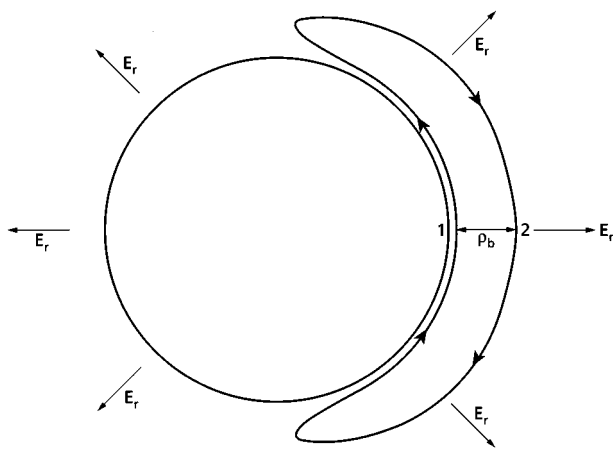


FIG. 1. Trapped particle orbit in poloidal cross section.

far from a magnetic surface in an orbit so that their change in kinetic energy, and therefore their orbit-averaged parallel velocity, is small. As a consequence their distribution function is essentially the equilibrium Maxwellian distribution, centered on $v_{\parallel} = 0$. This situation is shown by the dashed curves in Fig. 2, where it can be seen that a discontinuity exists in the distribution function at the trapped/passing boundary. In the low collision frequency regime, collisions are only important in a region of width $d_j \sim \sqrt{\nu_j / \epsilon(\omega + \epsilon^{1/2} k_{\parallel} v_{thj})}$ close to this boundary, where they smooth the distribution function

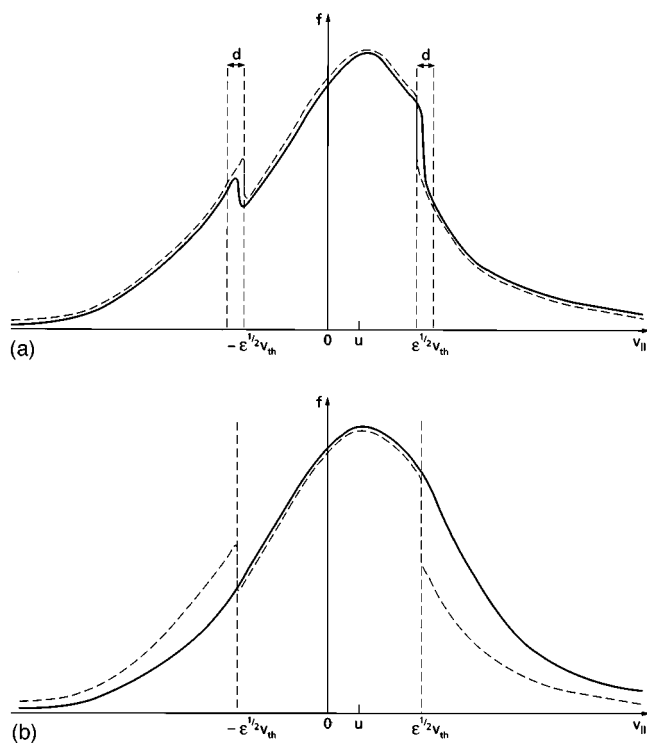


FIG. 2. Distribution function f in the (a) low collision frequency regime showing the “dissipation layer” where collisions resolve the discontinuity in the flux surface varying component, and (b) the fluid collision frequency regime, or the flux surface average of the function in the low collision frequency regime. In both cases the dashed curves show the distribution function in the total absence of collisions.

as shown by the full curve in Fig. 2(a). Thus, in the low collision frequency regime the dominant collisional dissipation results from the particles that exist in this narrow region of phase space. As this region is important for the determination of the propagation frequency, we shall discuss it in more detail shortly.

In nonlinear theory there is a further subtlety associated with the bootstrap current in the low collision frequency regime. Linear theory based on the above physics has been applied to trapped ion modes¹⁹ and the internal kink mode²⁰ and then the situation is as described above, with a net parallel fluid flow $u_{\parallel} \sim \epsilon^{3/2} u$ obtained from the distribution function of Fig. 2(a). If we were to introduce equilibrium gradients into the problem, then u would have additional terms proportional to these gradients, and the fluid flow would incorporate the $O(\epsilon^{3/2})$ diamagnetic flow associated with the banana orbits of the trapped particles, which provides the “seed” for the bootstrap current.²¹ In the linear, low collision frequency theory collisions are unable to transfer this trapped particle flow to the bulk of the passing particles, and therefore no significant perturbed bootstrap current is generated. However, in the nonlinear theory presented below, we find it is necessary to take into account the effect of the perturbations in the fields on the particle orbits. The mechanism is most straightforward to understand for the electrons, whose motion around the island is dominated by the parallel streaming. The ions are more complicated because their drifts across the island are comparable with their parallel streaming around it, but the result is the same. Thus, in a low collision frequency plasma an electron will make many transits around the island before experiencing a collision, and will average over local effects around the island between collisions. Collisions are therefore able to restore a Maxwellian velocity distribution in the *flux surface average* of the distribution around the island, which is then a shifted Maxwellian with a flow u , as shown in Fig. 2(b). However, this is not the case *locally*, and the dissipation layer discussed above still exists in the nonlinear theory. Thus, the nonlinear theory predicts $O(1)$ parallel flows that are constant on a perturbed flux surface in each particle species, and an $O(\epsilon^{3/2})$ flow that varies within a flux surface, but which flux-surface averages to zero. Therefore, even in the low collision frequency theory the standard $O(\epsilon^{1/2})$ bootstrap current perturbation is predicted, resulting from a difference between the ion and electron parallel flows. In contrast, in the fluid regime the particles experience many collisions as they travel around the magnetic island, but few during a transit time around the tokamak. Thus, in this case collisions force the distribution function averaged over a poloidal transit to take up the shifted Maxwellian. Then the distribution function does not exhibit the narrow dissipation layer of the collisionless theory and resembles that shown in Fig. 2(b) *locally*, i.e. an $O(1)$ flow $u_{\parallel} = u$ that varies around the flux surfaces of the magnetic island. This distinction between whether the flow is a flux surface quantity or not is important for determining the polarization current, as demonstrated below.

The neoclassical ion polarization current is enhanced over the FLR polarization current because of the large banana width relative to the ion Larmor radius. In the fluid

regime poloidal ion flows are strongly damped as a result of the high poloidal viscosity. As shown in Ref. 18 this implies a large radial current (a factor of order q^2/ϵ^2 greater than the FLR ion polarization current) must flow to maintain torque balance; this is the neoclassical polarization current. However, in a low collision frequency regime ion collisions are not so important, the poloidal ion viscosity does not dominate the poloidal force balance and the poloidal ion flow is not necessarily zero. This modifies the parallel ion flow, $u_{\parallel}$, and affects the standard neoclassical polarization current, which is related to the time variation of $u_{\parallel}$,

$$j_r \approx n \frac{m_i c}{B_{\theta}} \frac{du_{\parallel}}{dt}, \quad (2)$$

where

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \mathbf{v}_E \cdot \nabla, \quad (3)$$

and $\mathbf{v}_E$ is the $\mathbf{E} \times \mathbf{B}$ drift velocity. When the time variation associated with the island propagation frequency, ω , dominates its growth, the operator d/dt acting on a flux surface quantity is zero, as we shall now show. The mobile electrons readily flow along the perturbed field lines, so that to leading order the parallel electric field is zero. Thus, the electrostatic potential associated with the magnetic island must be a flux surface function in the frame of reference where the magnetic islands are stationary. Clearly, if the electrostatic potential is a flux surface quantity, then the operator $\mathbf{v}_E \cdot \nabla$ on a flux surface quantity must be zero. Furthermore, the operator d/dt is independent of the frame of reference, so that this result also holds in the frame where the islands propagate. In a fluid regime, where the $O(1)$ flow does vary around a flux surface, $du_{\parallel}/dt \sim \omega c E_r / B_{\theta}$, and one obtains the standard q^2/ϵ^2 neoclassical enhancement of the polarization current. However, in the collisionless case we have seen that to leading order the parallel flow is a flux surface quantity equal to the flux surface average of $u = c E_r / B_{\theta}$. As a result this leading order flow does not contribute to the ion polarization current; instead, this is generated by the smaller “seed” current of the trapped particles, yielding $du_{\parallel}/dt \sim \epsilon^{3/2} \omega c E_r / B_{\theta}$. Thus, in the low collision frequency regime the neoclassical enhancement of the polarization current is only $q^2/\epsilon^{1/2}$, despite the ion flow being $O(1)$, in accord with results obtained in linear theory.²⁰

We now return to the discussion of the “dissipation layer,” which exists for both species in the low collision frequency regime, $\nu_j < \epsilon \omega$. The ions are the most straightforward if we consider the limit $\omega < \epsilon^{-1/2} \rho_{\theta i}$, in which case $\omega \gg k_{\parallel} v_{ti}$. The dissipation layer then lies fully in the passing particle phase space and has a width given by $d_i \sim \sqrt{\nu_i / \omega}$, which corresponds to the balance of the effective collision frequency for scattering across the layer with the mode frequency. For the electrons $k_{\parallel} v_{te} \gg \omega$, and then two layers close to the trapped/passing boundary exist. In the passing region the layer width is given by $d_e^p \sim (\nu_e / k_{\parallel} v_{te})^{1/2}$. However, in the trapped region of phase space, $v_{\parallel e}$ averages to zero and a second layer exists whose width is given by $d_e^t \sim (\nu_e / \omega)^{1/2} \gg d_e^p$. These two situations are shown in Fig. 3.

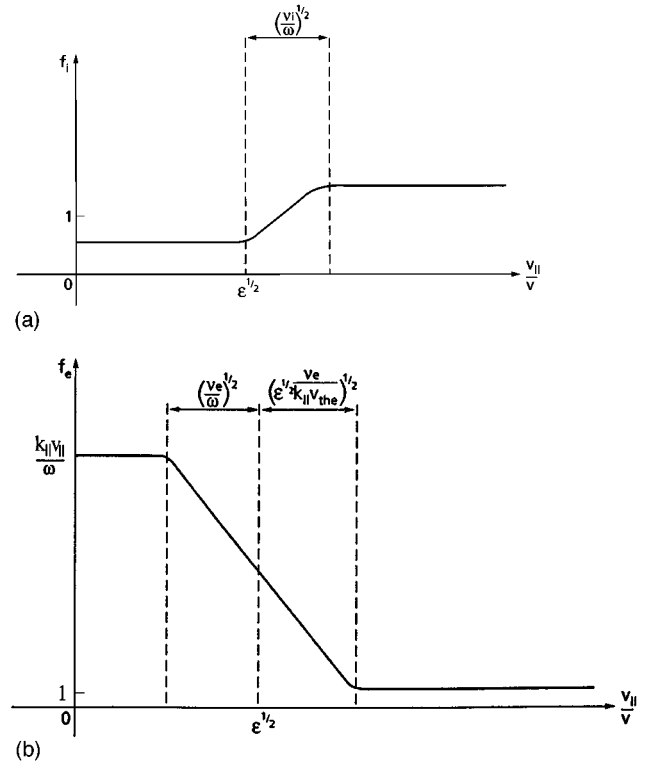


FIG. 3. Schematic diagram showing the “dissipation layers” close to the trapped/passing boundary for (a) ions and (b) electrons.

The propagation frequency is determined by a toroidal torque balance and results from currents that flow out of phase with the magnetic perturbation as a consequence of dissipation processes. For the ions, this current is carried by the barely passing particles resulting in a current $\sim (\nu_i / \omega)^{1/2}$. The case of the electrons is more subtle. Clearly, the trapped particles cannot carry a current and therefore the out-of-phase electron current must be carried by the barely passing electrons in the layer of width $\sim (\nu_e / k_{\parallel} v_{te})^{1/2}$. However, the electrons in the trapped dissipation layer enhance the current carried by the barely passing particles by collisional scattering across the trapped/passing boundary, so that the out-of-phase electron current is proportional to the trapped region layer width $d_e^t \sim (\nu_e / \omega)^{1/2}$. To justify this we note that the response of the electron distribution function, h_e , can be represented schematically as

$$(\omega - k_{\parallel} v_{\parallel}) h_e \sim S,$$

where the source S is the same order of magnitude as $k_{\parallel} v_{\parallel}$. In the passing region $k_{\parallel} v_{\parallel} \gg \omega$ and then $h_e \sim S / k_{\parallel} v_{\parallel}$. However, in the trapped region $v_{\parallel}$ averages to zero and $h_e \sim S / \omega \gg S / k_{\parallel} v_{\parallel}$. This situation is illustrated in Fig. 3(b) for $S \sim k_{\parallel} v_{\parallel}$. By linear interpolation across the two “dissipation layers,” we deduce that the value of h_e at the trapped/passing boundary is $\sim (\nu_e / k_{\parallel} v_{\parallel})^{1/2} (\nu_e / \omega)^{-1/2} (S / \omega) \sim S / (k_{\parallel} v_{\parallel})^{1/2}$. Taking account of the width of the dissipation layer in the passing region, we deduce that the electrons in this layer carry a current $\sim (\nu_e / k_{\parallel} v_{\parallel})^{1/2} S / (k_{\parallel} v_{\parallel})^{1/2}$. Now $S \sim k_{\parallel} v_{\parallel}$, so that the electron “layer current” is $\sim (\nu_e / \omega)^{1/2}$. Thus, the current carried by the barely passing electrons exceeds that of the ions by the factor $(\nu_e / \nu_i)^{1/2}$. This yields a torque on

the island that is proportional to $\nu_e^{1/2}$, with a constant of proportionality that depends on the propagation frequency ω . Torque balance requires that this be set to zero and therefore determines $\omega = \omega_{*e}(1 + \eta_e/4)$, independent of ν_e . However, the energy dependence of ν_e does affect the coefficient of η_e .

In this paper we develop the ideas discussed above to formally derive a neoclassical theory for a single helicity magnetic island in a low collision frequency plasma. For the present we neglect the effects of sheared plasma flows and any external interactions, with error fields for example, and we work in a frame of reference in which the equilibrium radial electric field is zero. We do not concern ourselves with an explicit calculation of the time evolution of the magnetic islands here, but concentrate instead on the conditions for a steady-state saturated solution. This is sufficient to derive the threshold width for island growth. The calculation proceeds as follows. In Sec. II A we introduce the magnetic geometry, set up the coordinate system, and derive the “dispersion relation,” which relates the current perturbation to the magnetic perturbation using Ampère’s law. In the following sections we then address the calculation of the current perturbation by deriving the particle distribution functions associated with the magnetic island using drift-kinetic equations for both species. Thus, in Secs. II B and II C we obtain expressions for the electron and ion responses, respectively, together with the self-consistent electrostatic potential, which ensures quasineutrality. These quantities are expressed in terms of three functions, h_e , h_i , and h , which are determined by constraint equations obtained and solved in Sec. II D. In Sec. II E we derive the full current perturbation and deduce the equation for the saturated island width. This is shown to have a threshold, provided the island propagation frequency lies within certain specified limits. On calculating the island propagation frequency in Sec. II F we find that the islands propagate in the electron diamagnetic direction and a threshold island width does indeed exist. Conclusions are drawn in Sec. III.

II. ISLAND DYNAMICS

A. Magnetic geometry and dispersion relation

A large aspect ratio, circular cross section, toroidal geometry is assumed. The equilibrium magnetic field is described by the poloidal flux χ and the toroidal angle ϕ :

$$\mathbf{B} = I(\chi) \nabla \phi + \nabla \phi \times \nabla \chi, \quad (4)$$

where $I(\chi) = RB_\phi$. With the poloidal angle θ we define the orthogonal coordinate system:

$$\nabla \phi \times \nabla \chi = r B_\theta \nabla \theta, \quad (5)$$

where B_θ is the poloidal component of the magnetic field. A single dominant helicity perturbation is imposed so that the total magnetic field is

$$\mathbf{B} = I(\chi) \nabla \phi + \nabla \phi \times \nabla (\chi + \psi), \quad (6)$$

with

$$\psi = \tilde{\psi} \cos \xi, \quad (7)$$

where ξ is a helical angle,

$$\xi = m \left(\theta - \frac{\phi}{q_s} \right) - \omega t. \quad (8)$$

Here m is the poloidal mode number, q_s is the value of the safety factor at the rational surface about which the perturbation is centered, and ω is the island propagation frequency. The subscript s denotes the fact that a quantity is to be evaluated at the rational surface. The perturbed flux ψ is related to a perturbation in the parallel component of the magnetic vector potential, $A_\parallel$:

$$\psi = -RA_\parallel. \quad (9)$$

For simplicity, a steady-state situation is assumed so that $\tilde{\psi}$ and ω are independent of time. Instead of the χ , ϕ , θ coordinate system introduced above, it is more convenient to work with the coordinates χ , θ , and ξ .

We introduce a flux surface function Ω , i.e., satisfying $\mathbf{B} \cdot \nabla \Omega = 0$,

$$\Omega = \frac{\psi_0(\chi)}{\tilde{\psi}} - \frac{\psi}{\tilde{\psi}}, \quad (10)$$

where

$$\psi_0 = 2\tilde{\psi} \frac{(\chi - \chi_s)^2}{w_\chi^2} \quad (11)$$

and

$$w_\chi^2 = \frac{4\tilde{\psi}q_s}{q'_s} \quad (12)$$

is related to the island half-width $w = w_\chi/(RB_\theta)$. Here a prime denotes a derivative with respect to χ . The surfaces of constant Ω then describe the magnetic island topology.

In the coordinate system above the parallel derivative operator is given by

$$\frac{\mathbf{B} \cdot \nabla}{B} \equiv \nabla_\parallel = \frac{1}{Rq} \frac{\partial}{\partial \theta} \bigg|_{\xi, \chi} + k_\parallel \frac{\partial}{\partial \xi} \bigg|_{\Omega, \theta}, \quad (13)$$

where we have written

$$k_\parallel = - \frac{m(\chi - \chi_s)}{Rq} \frac{q'_s}{q_s}. \quad (14)$$

The parallel derivative is annihilated by an integral operator defined in terms of the two angular averages:

$$\langle \cdots \rangle_\theta = \frac{1}{2\pi} \oint \cdots d\theta, \quad (15)$$

$$\langle \cdots \rangle_\Omega = \frac{\oint \cdots [\Omega + \cos \xi]^{-1/2} d\xi}{\oint [\Omega + \cos \xi]^{-1/2} d\xi}. \quad (16)$$

For passing particles, the annihilating operator is then

$$\langle \langle Rq \cdots \rangle_\theta \rangle_\Omega.$$

Using the parallel component of Ampère’s law integrated through the island region, where the current perturbation exists, and projecting out the $\cos \xi$ and $\sin \xi$ components in turn yields the nonlinear “dispersion relation” for the saturated island width w and propagation frequency ω ,

$$\sum_{\pm} \int_{-1}^{\infty} d\Omega \oint \frac{\bar{J}_{\parallel} \cos \xi}{\sqrt{\Omega + \cos \xi}} d\xi = \frac{c}{8\sqrt{2}} s\Delta' \frac{wB}{Rq}, \quad (17)$$

$$\sum_{\pm} \int_{-1}^{\infty} d\Omega \oint \frac{\bar{J}_{\parallel} \sin \xi}{\sqrt{\Omega + \cos \xi}} d\xi = 0. \quad (18)$$

Here the sum is over $\chi > \chi_s$ and $\chi < \chi_s$ regions, and $\bar{J}_{\parallel}$ is the θ average of the current perturbation $J_{\parallel}$. In deriving the second relation, which is equivalent to the toroidal torque balance, any external magnetic perturbation source has been neglected. This pair of equations can be solved for w and ω once the current perturbation, $J_{\parallel}$, associated with the magnetic island, is known. This is calculated from the electron and ion distribution functions, which are obtained in the following sections.

B. Electron response

The electron response to the magnetic perturbation defined in the previous section is described by the drift kinetic equation:²³

$$\begin{aligned} \frac{\partial f_j}{\partial t} + v_{\parallel} \nabla_{\parallel} f_j + \mathbf{v}_E \cdot \nabla f_j + \mathbf{v}_d \cdot \nabla f_j + \frac{q_j}{m_j} \frac{v_{\parallel} E_{\parallel}}{v} \frac{\partial f_j}{\partial v} \\ - \frac{q_j}{m_j} \frac{\mathbf{v}_d \cdot \nabla \Phi}{v} \frac{\partial f_j}{\partial v} = C(f_j), \end{aligned} \quad (19)$$

where $\mathbf{v}_E$ and $\mathbf{v}_d$ are the $\mathbf{E} \times \mathbf{B}$ and magnetic drifts, respectively. Here Φ is the electrostatic potential perturbation that arises because of the different responses of the electrons and ions and is to be obtained from quasineutrality, and C is the collision operator. Spatial derivatives are to be taken at constant magnetic moment $\mu = v_{\perp}^2/2B$ and kinetic energy $v^2/2$. Then the magnetic drift is $\mathbf{v}_d = -\mathbf{v}_{\perp} \mathbf{b} \times \nabla (v_{\parallel}/\omega_{ce})$, where $\mathbf{b}$ represents a unit vector in the magnetic field direction and $\omega_{cj} = q_j B/m_j c$. We express the distribution function as

$$f_e = \left(1 - \frac{q_e \Phi}{T_e}\right) F_{Me} + g_e; \quad (20)$$

F_{Me} is a Maxwellian distribution, taken to be a function of χ only, so that g_e is determined from the equation

$$\begin{aligned} \omega \frac{\partial g_e}{\partial \xi} \Big|_{\chi} - \frac{v_{\parallel}}{Rq} \frac{\partial g_e}{\partial \theta} - k_{\parallel} v_{\parallel} \frac{\partial g_e}{\partial \xi} \Big|_{\Omega} - \mathbf{v}_d \cdot \nabla g_e \\ - \frac{c(\mathbf{B} \times \nabla \Phi)}{B^2} \cdot \nabla g_e + \frac{q_e}{m_e} \frac{(\mathbf{v}_d \cdot \nabla \Phi)}{v} \frac{\partial g_e}{\partial v} + C(g_e) \\ = \frac{q_e F_{Me}}{T_e} \left[\left(\omega - \omega_{*e}^T \right) \left(\frac{\partial \Phi}{\partial \xi} \Big|_{\chi} - \frac{v_{\parallel}}{c} \frac{\partial A_{\parallel}}{\partial \xi} \right) - \frac{q_e \Phi}{T_e} \mathbf{v}_d \cdot \nabla \Phi \right] \\ + \left(1 - \frac{q_e \Phi}{T_e} \right) \frac{I v_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ce}} \right) \frac{F_{Me}}{n} \frac{dn}{d\chi} \frac{\omega_{*e}^T}{\omega_{*e}}, \end{aligned} \quad (21)$$

where, unless explicitly indicated, the partial derivatives are taken in the (χ, ξ, θ) coordinate system. Small terms to be

identified shortly have been neglected in Eq. (21). We have defined the diamagnetic frequency $\omega_{*j} = mcT_j n' / q_j q n$, where the prime denotes a derivative with respect to χ , and $\omega_{*j}^T = \omega_{*j} [1 + (v^2/v_{thj}^2 - \frac{3}{2}) \eta_j]$. It is convenient to work with velocity variables v and λ , where $\lambda = 2\mu/v^2$ is the pitch angle. In terms of these variables the parallel velocity and velocity space integral are

$$v_{\parallel} = \sigma v (1 - \lambda B)^{1/2}, \quad (22)$$

$$\int d^3 \mathbf{v} = \pi B \sum_{\sigma} \int_0^{\infty} v^2 dv \int_0^{B^{-1}} \frac{d\lambda}{(1 - \lambda B)^{1/2}},$$

where σ is the sign of $v_{\parallel}$; the trapped/passing boundary then corresponds to $\lambda = B_{\max}^{-1}$, the inverse of the maximum value of the magnetic field on a given equilibrium flux surface.

In order to solve the drift-kinetic equation for g_e , we define two small quantities, $\Delta = w/r$ and $\delta_e = \rho_{\theta e}/w$, where r is the minor radius of the rational surface. Anticipating the result that $E_{\parallel} \sim \Delta^2$ because of the leading-order cancellation between $\partial A_{\parallel}/\partial t$ and $\nabla_{\parallel} \Phi$, Eq. (21) is correct to $O(\Delta)$, where we assume the orderings

$$\frac{q_e \Phi}{T_e} \sim \Delta, \quad \frac{g_e}{F_{Me}} \sim \Delta, \quad \frac{\tilde{\Phi}}{\Phi} \sim \Delta, \quad k_{\theta} w \sim \Delta, \quad \nu_{*e} \leq \Delta, \quad (23)$$

with $\tilde{\Phi}$ the difference between Φ and its θ average. The collision frequency ordering is derived, assuming a low collisionality plasma formally satisfying $\nu_e \leq k_{\parallel} v_{the}$, though in Sec. II F we shall take it to be even smaller, $\nu_e < \epsilon \omega$. The terms in Eq. (21) are then of relative order:

$$\delta_e \Delta : 1 : \Delta : \delta_e : \delta_e \Delta : \delta_e \Delta : \delta_e \Delta : \delta_e \Delta : \delta_e \Delta.$$

We solve Eq. (21) by a double expansion in δ_e and Δ ; thus

$$g_e = \sum_{i,j} \delta_e^i \Delta^j g_e^{(i,j)}. \quad (24)$$

To order Δ^0 we have

$$-\frac{v_{\parallel}}{Rq} \frac{\partial g_e^{(i,0)}}{\partial \theta} - \mathbf{v}_d \cdot \nabla g_e^{(i,0)} - \frac{I v_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ce}} \right) \frac{\omega_{*e}^T}{\omega_{*e}} \frac{F_{Me}}{n} \frac{dn}{d\chi} = 0, \quad (25)$$

so that the leading-order expression for g_e satisfies

$$\frac{v_{\parallel}}{Rq} \frac{\partial g_e^{(0,0)}}{\partial \theta} = 0, \quad (26)$$

and we learn $g_e^{(0,0)} = \bar{g}_e^{(0,0)}$, where a bar over a quantity indicates that it is independent of θ , i.e. $\bar{g}_e = \bar{g}_e(\xi, \chi)$. The $O(\Delta^0 \delta_e)$ equation is

$$\begin{aligned} -\frac{v_{\parallel}}{Rq} \frac{\partial g_e^{(1,0)}}{\partial \theta} - \frac{I v_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ce}} \right) \frac{\partial \bar{g}_e^{(0,0)}}{\partial \chi} \\ - F_{Me} \frac{I v_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ce}} \right) \frac{\omega_{*e}^T}{\omega_{*e}} \frac{1}{n} \frac{dn}{d\chi} = 0. \end{aligned} \quad (27)$$

This can be integrated to yield

$$g_e^{(1,0)} = -\frac{I v_{\parallel}}{\omega_{ce}} \left(\frac{\partial \bar{g}_e^{(0,0)}}{\partial \chi} + \frac{F_{Me}}{n} \frac{dn}{d\chi} \frac{\omega_{*e}^T}{\omega_{*e}} \right) + \bar{h}_e, \quad (28)$$

where $\bar{h}_e$ is an, as yet, arbitrary function of ξ and χ , which will be determined by a solubility condition on the higher-order equations. We now proceed to the $O(\Delta)$ equation, which is

$$\begin{aligned} & \omega \left. \frac{\partial g_e^{(i,0)}}{\partial \xi} \right|_{\chi} - \frac{v_{\parallel}}{Rq} \frac{\partial g_e^{(i,1)}}{\partial \theta} - k_{\parallel} v_{\parallel} \left. \frac{\partial g_e^{(i,0)}}{\partial \xi} \right|_{\Omega} \\ & - c \frac{(\mathbf{B} \times \nabla \Phi^{(0)})}{B^2} \cdot \nabla g_e^{(i,0)} + \frac{q_e}{m_e} \frac{(\mathbf{v}_d \cdot \nabla \Phi)}{v} \frac{\partial g_e^{(i,0)}}{\partial v} \\ & - \mathbf{v}_d \cdot \nabla g_e^{(i,1)} - \mathbf{v}_d \cdot \nabla \xi \frac{\partial g_e^{(i,0)}}{\partial \xi} + C(g_e^{(i,0)}) \\ & = \frac{q_e F_{Me}}{T_e} \left[(\omega - \omega_{*e}^T) \left(\left. \frac{\partial \Phi}{\partial \xi} \right|_{\chi} - \frac{v_{\parallel}}{c} \frac{\partial A_{\parallel}}{\partial \xi} \right) - \frac{q_e \Phi}{T_e} \mathbf{v}_d \cdot \nabla \Phi \right] \\ & - \frac{q_e \Phi}{T_e} \frac{I v_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ce}} \right) \frac{\omega_{*e}^T}{\omega_{*e}} \frac{F_{Me}}{n} \frac{dn}{d\chi}. \end{aligned} \quad (29)$$

An equation for $\bar{g}_e^{(0,0)}$ results from the solubility condition for the $O(\Delta \delta_e^0)$ equation:

$$\begin{aligned} & - \frac{v_{\parallel}}{Rq} \frac{\partial g_e^{(0,1)}}{\partial \theta} - k_{\parallel} v_{\parallel} \left. \frac{\partial \bar{g}_e^{(0,0)}}{\partial \xi} \right|_{\Omega} + C(\bar{g}_e^{(0,0)}) \\ & = - \frac{q_e F_{Me}}{T_e} (\omega - \omega_{*e}^T) \frac{v_{\parallel}}{c} \frac{\partial A_{\parallel}}{\partial \xi}. \end{aligned} \quad (30)$$

For passing particles the solubility condition is derived by multiplying the above equation by $Rq/v_{\parallel}$, integrating over a period in θ and requiring that $g_e^{(0,1)}$ be periodic in θ . Thus, for passing particles we have

$$\begin{aligned} & - Rq k_{\parallel} \left. \frac{\partial \bar{g}_e^{(0,0)}}{\partial \xi} \right|_{\Omega} + \left\langle \frac{Rq}{v_{\parallel}} C(\bar{g}_e^{(0,0)}) \right\rangle_{\theta} \\ & = - \frac{q_e F_{Me}}{T_e} \frac{Rq}{c} (\omega - \omega_{*e}^T) \frac{\partial A_{\parallel}}{\partial \xi}. \end{aligned} \quad (31)$$

Using the result

$$k_{\parallel} \left. \frac{\partial \chi}{\partial \xi} \right|_{\Omega} = \frac{m}{q} \frac{\partial A_{\parallel}}{\partial \xi}, \quad (32)$$

we then find the leading-order distribution function in the passing region of phase-space is

$$\bar{g}_e^{(0,0)} = \frac{qq_e F_{Me}}{mc T_e} (\omega - \omega_{*e}^T) [\chi - h(\Omega)]. \quad (33)$$

At this point $h(\Omega)$ is an arbitrary flux surface function resulting as a constant of integration. We shall see later that its form follows from consideration of the radial transport in the vicinity of the magnetic island.

To determine $\bar{g}_e^{(0,0)}$ in the trapped region we multiply Eq. (30) by $Rq/|v_{\parallel}|$, sum over σ , and integrate between bounce points (defined by $v_{\parallel}=0$). The condition on $g_e^{(0,1)}$ is that $g_e^{(0,1)}(\theta = \pm \theta_b, \sigma=1) = g_e^{(0,1)}(\theta = \pm \theta_b, \sigma=-1)$, representing continuity at a bounce point. The same condition must also hold for $\bar{g}_e^{(0,0)}$ in the trapped region, but as this is

independent of θ we know that $\bar{g}_e^{(0,0)}$ is in fact independent of σ . This result is used to calculate the trapped particle constraint equation for $\bar{g}_e^{(0,0)}$,

$$\sum_{\sigma=\pm 1} \left\langle \frac{Rq}{|v_{\parallel}|} C(\bar{g}_e^{(0,0)}) \right\rangle_{\theta} = 0. \quad (34)$$

This forces continuity in pitch angle, so that $\bar{g}_e^{(0,0)}$ is given by Eq. (33) in both the trapped and passing regions.

Particle conservation at the bounce points means that $\bar{h}_e$ must be independent of σ in the trapped region of phase space; no such constraint is applicable in the passing region so h_e may depend upon σ there. The precise form of h_e is calculated from the $O(\Delta \delta_e)$ equation, and this involves the electrostatic potential perturbation, Φ . Here Φ is determined using quasineutrality from knowledge of the leading-order ion response; this is calculated in the next section.

C. Ion response

The ion Larmor radius is also assumed to be much smaller than the island width so that its response to the magnetic perturbation is described by the drift kinetic equation, Eq. (19), as well. The distribution function is written as

$$f_i = \left(1 - \frac{q_i \Phi}{T_i} \right) F_{Mi} + g_i, \quad (35)$$

where F_{Mi} is a Maxwellian distribution, taken to be a function of χ only, and g_i satisfies the equation

$$\begin{aligned} & \omega \left. \frac{\partial g_i}{\partial \xi} \right|_{\chi} - \frac{v_{\parallel}}{Rq} \frac{\partial g_i}{\partial \theta} - k_{\parallel} v_{\parallel} \left. \frac{\partial g_i}{\partial \xi} \right|_{\Omega} - \mathbf{v}_d \cdot \nabla g_i - \frac{c(\mathbf{B} \times \nabla \Phi)}{B^2} \cdot \nabla g_i \\ & + \frac{q_i}{m_i} \frac{(\mathbf{v}_d \cdot \nabla \Phi)}{v} \frac{\partial g_i}{\partial v} + C(g_i) \\ & = \frac{q_i F_{Mi}}{T_i} \left[(\omega - \omega_{*i}^T) \left(\left. \frac{\partial \Phi}{\partial \xi} \right|_{\chi} - \frac{v_{\parallel}}{c} \frac{\partial A_{\parallel}}{\partial \xi} \right) - \frac{q_i \Phi}{T_i} \mathbf{v}_d \cdot \nabla \Phi \right] \\ & + \left(1 - \frac{q_i \Phi}{T_i} \right) \frac{I v_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ci}} \right) \frac{F_{Mi}}{n} \frac{dn}{d\chi} \frac{\omega_{*i}^T}{\omega_{*i}}. \end{aligned} \quad (36)$$

For the ions we consider $\rho_{\theta i}/w \sim 1$. However, making use of the small inverse aspect ratio, ϵ , we define the small parameter $\delta_i = \epsilon^{1/2} \rho_{\theta i}/w$ and perform a double expansion in δ_i and Δ . Thus, we have the orderings

$$\frac{q_i \Phi}{T_i} \sim \Delta, \quad \frac{g_i}{F_{Mi}} \sim \Delta, \quad \frac{\tilde{\Phi}}{\Phi} \sim \Delta, \quad \frac{v_i}{\omega} \leq 1, \quad (37)$$

where the collision frequency ordering is derived assuming a low collisionality plasma satisfying $\nu_i < \omega_{*}$. The consecutive terms in Eq. (36) are then of relative order:

$$\Delta : 1 : \Delta : \delta_i : \Delta : \delta_i \Delta : \Delta : \delta_i \Delta : \delta_i : \delta_i \Delta,$$

and we expand

$$g_i = \sum_{k,j} \delta_i^k \Delta^j g_i^{(k,j)}. \quad (38)$$

The $O(\Delta^0)$ equation is

$$-\frac{v_{\parallel}}{Rq} \frac{\partial g_i^{(k,0)}}{\partial \theta} - \frac{Iv_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ci}} \right) \frac{\partial g_i^{(k,0)}}{\partial \chi} \\ = \frac{Iv_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ci}} \right) \frac{\omega_{*i}^T F_{Mi}}{\omega_{*i} n} \frac{dn}{d\chi}, \quad (39)$$

so that to $O(\Delta^0 \delta_i^0)$ we have

$$\frac{\partial g_i^{(0,0)}}{\partial \theta} = 0, \quad (40)$$

and $g_i^{(0,0)} = \bar{g}_i^{(0,0)}$, independent of θ . As discussed in the previous section for the electrons, this implies that $\bar{g}_i^{(0,0)}$ is independent of σ in the trapped region of phase space. The $O(\Delta^0 \delta_i)$ equation is

$$-\frac{v_{\parallel}}{Rq} \frac{\partial g_i^{(1,0)}}{\partial \theta} - \frac{Iv_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ci}} \right) \frac{\partial \bar{g}_i^{(0,0)}}{\partial \chi} \\ = \frac{Iv_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ci}} \right) \frac{F_{Mi}}{n} \frac{dn}{d\chi} \frac{\omega_{*i}^T}{\omega_{*i}}, \quad (41)$$

so that

$$g_i^{(1,0)} = -\frac{Iv_{\parallel}}{\omega_{ci}} \left(\frac{\partial \bar{g}_i^{(0,0)}}{\partial \chi} + \frac{F_{Mi}}{n} \frac{dn}{d\chi} \frac{\omega_{*i}^T}{\omega_{*i}} \right) + \bar{h}_i. \quad (42)$$

Proceeding to the $O(\Delta)$ equation, we obtain

$$\omega \frac{\partial g_i^{(k,0)}}{\partial \xi} \Big|_{\chi} - \frac{v_{\parallel}}{Rq} \frac{\partial g_i^{(k,1)}}{\partial \theta} - k_{\parallel} v_{\parallel} \frac{\partial g_i^{(k,0)}}{\partial \xi} \Big|_{\Omega} - \mathbf{v}_E \cdot \nabla g_i^{(k,0)} \\ - \frac{Iv_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ci}} \right) \frac{\partial g_i^{(k,1)}}{\partial \chi} - \frac{Iv_{\parallel}}{Rq} \frac{\partial}{\partial \chi} \left(\frac{v_{\parallel}}{\omega_{ci}} \right) \\ \times (\mathbf{B} \times \nabla \chi) \cdot \nabla g_i^{(k,0)} + \frac{q_i}{m_i} \frac{(\mathbf{v}_d \cdot \nabla \Phi)}{v} \frac{\partial g_i^{(k,0)}}{\partial v} + C(g_i^{(k,0)}) \\ = -\frac{q_i \Phi}{T_i} \frac{Iv_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ci}} \right) \frac{\omega_{*i}^T F_{Mi}}{\omega_{*i} n} \frac{dn}{d\chi} + \frac{q_i F_{Mi}}{T_i} \\ \times \left[(\omega - \omega_{*i}^T) \left(\frac{\partial \Phi}{\partial \xi} \Big|_{\chi} - \frac{v_{\parallel}}{c} \frac{\partial A_{\parallel}}{\partial \xi} \right) \right. \\ \left. - \frac{q_i \Phi}{T_i} \frac{Iv_{\parallel}}{Rq} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{ci}} \right) \frac{\partial \Phi}{\partial \chi} \right]. \quad (43)$$

The $O(\Delta \delta_i^0)$ equation is then

$$\omega \frac{\partial g_i^{(0,0)}}{\partial \xi} \Big|_{\chi} - \frac{v_{\parallel}}{Rq} \frac{\partial g_i^{(0,1)}}{\partial \theta} - k_{\parallel} v_{\parallel} \frac{\partial g_i^{(0,0)}}{\partial \xi} \Big|_{\Omega} - \mathbf{v}_E \cdot \nabla g_i^{(0,0)} \\ + C(g_i^{(0,0)}) = \frac{q_i F_{Mi}}{T_i} (\omega - \omega_{*i}^T) \left(\frac{\partial \Phi}{\partial \xi} \Big|_{\chi} - \frac{v_{\parallel}}{c} \frac{\partial A_{\parallel}}{\partial \xi} \right). \quad (44)$$

We annihilate the term in $g_i^{(0,1)}$ in the passing region of phase space by again multiplying by $Rq/v_{\parallel}$ and integrating over a period in θ . The resulting equation yields an expression for $g_i^{(0,0)}$ in terms of the electrostatic potential, Φ , which will be determined using quasineutrality. A self-consistent solution has $E_{\parallel} = 0$ to leading order, which represents the freedom of the mobile electrons to flow rapidly

along the oscillating magnetic field lines of the island to short out the parallel electric field. We will find that by integrating $E_{\parallel} = 0$ along a field line to give

$$\Phi = \frac{\omega q}{mc} [\chi - h(\Omega)], \quad (45)$$

and choosing $h(\Omega)$, an arbitrary function of integration, to be the same as that introduced into the electron response, Eq. (33), we can satisfy the quasineutrality requirement. Using Eq. (45) it is straightforward to show

$$\frac{d}{dt} \equiv \frac{\partial}{\partial t} + \mathbf{v}_E \cdot \nabla \\ = \frac{dh}{d\Omega} \frac{\omega}{m \tilde{\psi}} Rq k_{\parallel} \frac{\partial}{\partial \xi} \Big|_{\Omega} + \frac{\omega}{m} \left(1 - \frac{dh}{d\Omega} \frac{1}{\tilde{\psi}} \frac{d\psi_0}{d\chi} \right) \frac{\partial}{\partial \theta}, \quad (46)$$

so that the $O(\Delta \delta_i^0)$ equation simplifies to

$$-Rk_{\parallel} q \left(\frac{dh}{d\Omega} \frac{\omega}{m \tilde{\psi}} + \frac{v_{\parallel}}{Rq} \right) \frac{\partial g_i^{(0,0)}}{\partial \xi} \Big|_{\Omega} - \frac{v_{\parallel}}{Rq} \frac{\partial g_i^{(0,1)}}{\partial \theta} + C(g_i^{(0,0)}) \\ = m \frac{(\omega - \omega_{*i}^T)}{\omega_{*i}} \left(\frac{dh}{d\Omega} \frac{\omega}{\tilde{\psi} m} + \frac{v_{\parallel}}{Rq} \right) \frac{\partial \psi}{\partial \xi} \frac{F_{Mi}}{n} \frac{dn}{d\chi}. \quad (47)$$

Annihilating the term in $g_i^{(0,1)}$, we find

$$g_i^{(0,0)} = \frac{(\omega - \omega_{*i}^T)}{\omega_{*i}} \frac{F_{Mi}}{n} \frac{dn}{d\chi} [\chi - h(\Omega)], \quad (48)$$

where the arbitrary function of Ω has again been chosen to be $h(\Omega)$ so that quasineutrality is satisfied. With this choice, it is straightforward to show that $g_i^{(0,0)}$, together with the solutions for Φ and $g_e^{(0,0)}$, are consistent with quasineutrality. This same solution also satisfies the trapped particle constraint equation.

D. Constraint equations

In the previous two sections we have determined the electron and ion distribution functions in terms of three arbitrary functions: $h(\Omega)$, $h_e(\xi, \chi)$, and $h_i(\xi, \chi)$. In this section we derive the constraint equations that determine these functions, and thereby deduce the full particle response.

We begin with the determination of the function $h(\Omega)$, following similar lines to those developed in Refs. 2 and 22. Here $h(\Omega)$ was originally introduced as an arbitrary function of integration and, in fact, it cannot be determined from the drift-kinetic equation as it stands. This is because any arbitrary function of Ω satisfies the drift-kinetic equation if $E_{\parallel} = 0$; therefore extra physics must be included to determine the form for $h(\Omega)$. The significance of $h(\Omega)$ can be illustrated by considering the electron density, which is obtained by integrating the distribution function over velocity space. Thus

$$n_e = n(\chi) - \frac{dn}{d\chi} \Big|_{\chi=\chi_s} [\chi - h(\Omega)], \quad (49)$$

where the first term results from the leading-order Maxwellian distribution and the remainder is the perturbation. Taylor expanding $n(\chi)$ about the rational surface, $\chi = \chi_s$, we then obtain

$$n_e = n_s + n'_s h(\Omega). \quad (50)$$

Therefore $h(\Omega)$ represents the density gradient in the vicinity of the island, and can only be determined when the effect of the island on the radial transport is known. To model the effect on radial transport we perturbatively introduce a term,

$$\frac{\partial}{\partial \chi} \Gamma_\chi$$

into the drift kinetic equation, where $\Gamma_\chi = -D \partial n / \partial \chi$ represents the radial particle flux and D is a diffusion coefficient that is assumed to vary slowly across the island. (Note that Ref. 16 considered the opposite limit, where the transport in the island region is dominated by the radial transport; that situation is relevant for small island widths.) The introduction of this new term results, in the absence of sources and sinks, in a solubility condition,

$$\left\langle \frac{\partial \Gamma_\chi}{\partial \chi} \right\rangle_\Omega = - \left\langle \frac{\partial}{\partial \chi} D \frac{\partial n}{\partial \chi} \right\rangle_\Omega = 0. \quad (51)$$

In terms of $h(\Omega)$, this constraint becomes

$$\left\langle \frac{\partial^2 h}{\partial \chi^2} \right\rangle_\Omega = 0, \quad (52)$$

which has the solution

$$\frac{dh}{d\Omega} = \frac{A}{Q}, \quad (53)$$

where

$$Q = \frac{1}{2\pi} \oint \sqrt{\Omega + \cos \xi} d\xi, \quad (54)$$

and A is a constant of integration to be determined by the boundary conditions, which differ inside and outside the magnetic island. Outside the island, defined by $\Omega > 1$, A is determined by the boundary condition

$$\lim_{\chi \rightarrow \pm \infty} h = \chi,$$

so that h is odd in χ . Since $h(\Omega)$ must be single valued at the rational surface, $\chi = \chi_s$, this implies it must in fact be zero there. Together with the fact that h is a flux surface quantity, this imposes the condition $h(\Omega) = 0$ inside the magnetic island. Matching the solutions at the island separatrix, we obtain

$$h(\Omega) = \Theta(\Omega - 1) \frac{w_\chi}{2\sqrt{2}} \int_1^\Omega \frac{d\Omega}{Q}, \quad (55)$$

where Θ is the Heaviside function and w_χ is defined to have the same sign as $(\chi - \chi_s)$.

We now proceed to derive expressions for $\bar{h}_e$ and $\bar{h}_i$. The derivation of these terms requires a different treatment in each of the collision frequency regimes, as we shall now show. We first consider the ion term, for which the constraint

equation is derived from the $O(\Delta \delta_i)$ contribution to Eq. (43). To obtain an expression for h_i in the passing region, we multiply the $O(\Delta \delta_i)$ contribution to Eq. (43) by $Rq/v_\parallel$ and integrate over a period in θ yet again. Using the fact that $g_i^{(0,1)}$ is independent of θ we obtain the following constraint equation:

$$-Rqk_\parallel \left\langle \left(1 + \frac{Rq}{v_\parallel} \frac{\omega}{m\tilde{\psi}} \frac{dh}{d\Omega} \right) \frac{\partial g_i^{(1,0)}}{\partial \xi} \right\rangle_\Omega \Big|_\theta + \left\langle \frac{Rq}{v_\parallel} C(g_i^{(1,0)}) \right\rangle_\theta - \left\langle \frac{Rq}{v_\parallel} \mathbf{v}_d \cdot \nabla \xi \right\rangle_\theta \frac{\partial g_i^{(0,0)}}{\partial \xi} \Big|_\chi = 0. \quad (56)$$

Substituting the expression in Eq. (48) for $g_i^{(0,0)}$, we then obtain

$$-Rqk_\parallel \left\langle \left(1 + \frac{Rq}{v_\parallel} \frac{\omega}{m\tilde{\psi}} \frac{dh}{d\Omega} \right) \frac{\partial g_i^{(1,0)}}{\partial \xi} \right\rangle_\Omega \Big|_\theta + \frac{I}{\tilde{\psi}} \frac{F_{Mi}}{n} \frac{dn}{d\chi} \frac{dh}{d\Omega} \frac{(\omega - \omega_{*i}^T)}{\omega_{*i}} \left\langle \frac{\partial}{\partial \chi} \left(\frac{v_\parallel}{\omega_{ci}} \right) \frac{\partial \chi}{\partial \xi} \right\rangle_\Omega \Big|_\theta + \left\langle \frac{Rq}{v_\parallel} C(g_i^{(1,0)}) \right\rangle_\theta = 0. \quad (57)$$

Before solving Eq. (57) explicitly, we discuss some of its consequences qualitatively. This equation encompasses both the collisional and the collisionless limits. In the collisional limit, the term in curly brackets is assumed to be small and then collisions alone determine the constraint equation for h_i . This yields the standard expressions for bootstrap current and the neoclassically enhanced ion polarization current¹⁸ (which is driven by the parallel flow). On the other hand, in the collisionless limit the term in curly brackets dominates, except in the neighborhood of the trapped/passing boundary, as we shall demonstrate shortly. However, using the operator defined in Eq. (16) to flux surface average Eq. (57) around the island, imposes a further condition on h_i ; this constraint is equivalent to the flux surface average of the collisional constraint and is crucial: it is this flux surface average constraint that leads to the bootstrap current. The point is that the term in curly brackets can only determine h_i to within an arbitrary flux surface function, $H_i(\Omega)$. Thus, if $H_i(\Omega)$ is defined so that the part of h_i that varies around a flux surface averages to zero around that flux surface, then $H_i(\Omega)$ in the collisionless theory satisfies the flux surface average of the constraint on h_i which exists in the collisional theory. As a result, the parallel flows in the collisionless theory are the flux surface averages of those in the collisional theory, but with an $O(\epsilon^{3/2})$ correction that varies around a flux surface. Thus, we shall find that the bootstrap current perturbation is independent of the collision frequency regime, whereas the ion polarization current, which is driven by a parallel derivative of the parallel flow, is an order $\epsilon^{3/2}$ smaller in the collisionless regime. Nevertheless, this ‘‘collisionless’’ neoclassical polarization current is still a factor $q^2/\epsilon^{1/2}$ larger than the FLR polarization current.

We now demonstrate the results described qualitatively above in a more quantitative manner. Treating collisions as small, we find that in the passing region h_i has the solution

$$\begin{aligned} \bar{h}_i = & -\frac{4I}{w_\chi^2} \frac{dh}{d\Omega} \frac{F_{Mi}}{n} \frac{dn}{d\chi} \frac{(\omega - \omega_{*i}^T)}{\omega_{*i}} \\ & \times \left[\left\langle \left(1 + \frac{Rq}{v_\parallel} \frac{\omega}{m\tilde{\psi}} \frac{dh}{d\Omega} \right) \frac{v_\parallel}{\omega_{ci}} \right\rangle_\theta + \frac{q_s}{q_s'} \frac{\partial}{\partial \chi} \left\langle \frac{v_\parallel}{\omega_{ci}} \right\rangle_\theta \right] \\ & \times \left\langle 1 + \frac{Rq}{v_\parallel} \frac{\omega}{m\tilde{\psi}} \frac{dh}{d\Omega} \right\rangle_\theta^{-1} (\chi - \langle \chi \rangle_\Omega) + H_i(\Omega). \end{aligned} \quad (58)$$

where $H_i(\Omega)$ is to be determined by the condition

$$\left\langle \left\langle \frac{Rq}{v_\parallel} C(g_i^{(1,0)}) \right\rangle_\theta \right\rangle_\Omega = 0. \quad (59)$$

The term in square brackets is only important for the determination of the polarization current, which we shall find is not important for $w \gtrsim \rho_{\theta i}$. The region $w \lesssim \rho_{\theta i}$ corresponds to $\omega/k_\parallel v_{\parallel i} > 1$, which is used to simplify Eq. (58).

A constraint equation for the trapped particles is obtained by multiplying the $O(\Delta\delta_i)$ contribution to Eq. (43) by $Rq/|v_\parallel|$, summing over $\sigma = \pm 1$, and integrating over θ between bounce points. For $\omega \gg k_\parallel v_{\parallel i}$ the result is

$$\frac{\omega}{m\tilde{\psi}} \frac{dh}{d\Omega} \left\langle \frac{Rq}{|v_\parallel|} \right\rangle_\theta k_\parallel \frac{\partial \bar{h}_i}{\partial \xi} \Big|_\Omega - \left\langle \frac{Rq}{|v_\parallel|} C(\bar{h}_i) \right\rangle_\theta = 0, \quad (60)$$

so that away from the trapped/passing boundary collisions are small and $h_i = H_i(\Omega)$, where

$$\left\langle \left\langle \frac{Rq}{|v_\parallel|} C(H_i) \right\rangle_\theta \right\rangle_\Omega = 0. \quad (61)$$

It can be seen that the passing particle solution for $\bar{h}_i$ approaches zero very rapidly at the trapped/passing boundary because of the logarithmic divergence in the average $\langle Rq/v_\parallel \rangle_\theta$. Thus, for all intents and purposes h_i is discontinuous at the boundary, as indicated by the dashed curve of Fig. 2. In this region collisions are important and tend to smooth the distribution function across the trapped/passing boundary. To estimate the region of phase space over which collisions are important, we note that the collision operator will be dominated by the pitch angle scattering in this region. Thus, the effective collision frequency is $\nu_{\text{eff}} \sim \nu_i v_{\text{th}i}^2 / (\Delta v_\parallel)^2$, where $\Delta v_\parallel$ is the width of the layer where collisions are important. Within this layer $\nu_{\text{eff}} \gtrsim \omega$, so that $\Delta v_\parallel / v_{\text{th}i} \sim (\nu_i / \epsilon \omega)^{1/2}$. This layer width is small, and therefore makes a negligible contribution to that piece of the current perturbation that is in phase with the magnetic perturbation. However, it is the only region where dissipation operates and provides the only source of an out-of-phase contribution to the current perturbation. It is therefore important in the torque balance relation that determines ω ; we shall return to calculate the form of this layer solution in Sec. II F when we address the island frequency. The above form for h_i is sufficient for a calculation of the in-phase contribution of the current perturbation (i.e., the $\cos \xi$ component).

In order to completely determine $\bar{h}_i$ we must calculate $H_i(\Omega)$ using the constraints in Eqs. (59) and (61). The term in h_i in Eq. (58) with the square bracketed factor does not contribute to Eq. (59) as its flux surface averages to zero. Therefore, $H_i(\Omega)$ is just the flux surface average of the expression for h_i , which would be calculated in the fluid regime. To calculate an explicit expression for $H_i(\Omega)$ we must consider a particular form for the collision operator. We choose a relatively simple momentum-conserving operator, which is expected to give good qualitative results, though numerical coefficients on temperature gradients may be in error. This model collision operator is²³

$$\begin{aligned} C_{ii}(g) = & \nu_{ii}(v) \left[2 \frac{(1-\lambda B)^{1/2}}{B} \frac{\partial}{\partial \lambda} \left(\lambda (1-\lambda B)^{1/2} \frac{\partial g}{\partial \lambda} \right) \right. \\ & \left. + \frac{v_\parallel \bar{u}_{\parallel i}}{v_{\text{th}i}^2} F_{Mi} \right], \\ C_{ee}(g) = & \nu_{ee}(v) \left[2 \frac{(1-\lambda B)^{1/2}}{B} \frac{\partial}{\partial \lambda} \left(\lambda (1-\lambda B)^{1/2} \frac{\partial g}{\partial \lambda} \right) \right. \\ & \left. + \frac{v_\parallel \bar{u}_{\parallel e}}{v_{\text{the}}^2} F_{Me} \right], \\ C_{ei}(g) = & \nu_{ei}(v) \left[2 \frac{(1-\lambda B)^{1/2}}{B} \frac{\partial}{\partial \lambda} \left(\lambda (1-\lambda B)^{1/2} \frac{\partial g}{\partial \lambda} \right) \right. \\ & \left. + \frac{2}{v_{\text{the}}^2} v_\parallel u_{\parallel i} F_{Me} \right], \end{aligned} \quad (62)$$

where $\nu_{ij}(v) = \nu_{ij}(v_{\text{th}})(v_{\text{th}}/v)^3$ and

$$\bar{u}_{\parallel j} = \frac{3\sqrt{\pi}}{2n} v_{\text{th}j}^3 \int d^3\mathbf{v} \frac{v_\parallel g_j}{v^3}, \quad (63)$$

$$u_{\parallel j} = \frac{1}{n} \int d^3\mathbf{v} v_\parallel g_j. \quad (64)$$

Ion–electron collisions are small and have been neglected.

We determine $H_i(\Omega)$ by first solving the collisional constraint for the auxiliary quantity, $\tilde{H}_i$:

$$\left\langle \frac{Rq}{v_\parallel} C_{ii}(g_i^{(1,0)}) \right\rangle_\theta = 0, \quad (65)$$

and then using the result $H_i(\Omega) = \langle \tilde{H}_i \rangle_\Omega$ in the collisionless regime. After some algebra we find

$$\tilde{H}_i = \frac{-\sigma v}{2} \left(A_i + \frac{\langle B \bar{u}_{\parallel i} \rangle_\theta}{v_{\text{th}i}^2} F_{Mi} \right) \int_{\lambda_c}^{\lambda} \frac{d\lambda}{\sqrt{1-\lambda B}} \Theta(\lambda_c - \lambda), \quad (66)$$

where $\lambda_c = B_{\text{max}}^{-1}$ is the inverse of the maximum value of the magnetic field on the flux surface and

$$A_i = \frac{Im_i c}{q_i} \left(\frac{\partial g_i^{(0,0)}}{\partial \chi} + \frac{F_{Mi}}{n} \frac{dn}{d\chi} \frac{\omega_{*i}^T}{\omega_{*i}} \right). \quad (67)$$

The “flow” $\bar{u}_{\parallel i}$ is constructed according to the definition given in Eq. (63), with the result

$$\frac{\langle B \bar{u}_{\parallel i} \rangle_\theta}{B} = \frac{\rho_{\theta i}}{L_n} v_{\text{th}i} \left\{ \left[\frac{\omega}{\omega_{*i}} - \left(1 - \frac{\eta_i}{2} \right) \right] \frac{\partial h}{\partial \chi} - \frac{\omega}{\omega_{*i}} \right\}, \quad (68)$$

where we have substituted for $g_i^{(0,0)}$ from Eq. (48). This result can be used in the expression for $\tilde{H}_i$, which is then flux surface averaged to derive $H_i(\Omega)$. For the moment, we neglect the contribution to the parallel flow which results from the difference between h_i and $H_i(\Omega)$, and which flux surface averages to zero (however, this part of the flow is important for the calculation of the polarization current, and we shall return to it in the following section). Then we have

$$u_{\parallel i} = \frac{1}{2} \frac{\rho_{\theta i}}{L_n} v_{\text{thi}} \left[\left(\frac{\omega}{\omega_{*i}} - (1 + \eta_i) \right) \frac{\partial h}{\partial \chi} - \frac{\omega}{\omega_{*i}} \right] + \frac{9}{8} K B^2 \eta_i \left\langle \frac{\partial h}{\partial \chi} \right\rangle_{\Omega}, \quad (69)$$

where we have defined

$$K = \int_0^{\lambda_c} \frac{\lambda d\lambda}{\langle \sqrt{1 - \lambda B} \rangle_{\theta}}. \quad (70)$$

The last term in Eq. (69) represents the effect of the poloidal flow damping and is a flux surface averaged quantity in this collisionless theory (this is not the case in the fluid theory).

Having determined the ion flow [the only consequence of $H_i(\Omega)$ that we are interested in here], we now consider the calculation of h_e , which is similar to the calculation for h_i described above. The constraint equation that determines h_e in the passing region is derived from the $O(\Delta \delta_e)$ contribution to Eq. (29) and has similar characteristics to the corresponding ion equation. It is

$$Rqk_{\parallel} \left\langle \left\langle \frac{\partial g_e^{(1,0)}}{\partial \xi} \right\rangle_{\Omega} \right\rangle_{\theta} + \frac{I F_{Me}}{\tilde{\psi}} \frac{dn}{n} \frac{dh}{d\chi} \frac{(\omega - \omega_{*e}^T)}{\omega_{*e}} \left\langle \frac{\partial}{\partial \chi} \left(\frac{v_{\parallel}}{\omega_{ce}} \right) \right\rangle_{\theta} \frac{\partial \chi}{\partial \xi} \Big|_{\Omega} - \left\langle \frac{Rq}{v_{\parallel}} C_e(g_e^{(1,0)}) \right\rangle_{\theta} = 0, \quad (71)$$

differing only because $\omega/k_{\parallel}v_{\parallel}$ is small for the electrons. As with the solution of the ion equation, electron collisions are not important except in a narrow layer close to the trapped/passing boundary; again, this gives a negligible contribution to the electron parallel flow in phase with the magnetic perturbation. However, as with the ions, this region is important in determining the out-of-phase contribution to the parallel flow and is therefore crucial in deriving the island propagation frequency. This layer region will be described in Sec. II F, but for the moment we are concerned only with the in-phase component of the electron parallel flow. Thus, following the ion calculation we find that in the passing region the electron function satisfies

$$\bar{h}_e = -\frac{4}{w_{\chi}^2} I \frac{F_{Me}}{n} \frac{dn}{d\chi} \frac{dh}{d\Omega} \frac{(\omega - \omega_{*e}^T)}{\omega_{*e}} \left\langle \left\langle \frac{v_{\parallel}}{\omega_{ce}} \right\rangle_{\theta} \right\rangle_{\Omega} + \frac{q_s}{q_s'} \frac{\partial}{\partial \chi} \left\langle \frac{v_{\parallel}}{\omega_{ce}} \right\rangle_{\theta} (\chi - \langle \chi \rangle_{\Omega}) + H_e(\Omega), \quad (72)$$

where $H_e(\Omega)$ satisfies the condition

$$\left\langle \left\langle \frac{Rq}{v_{\parallel}} C_e(g_e^{(1,0)}) \right\rangle_{\theta} \right\rangle_{\Omega} = 0. \quad (73)$$

The trapped particle dynamics are only important in the dissipation layers and we shall discuss these in Sec. II F. For the in-phase contribution to the current perturbation the dominant piece comes from the passing particle distribution function, which is determined by Eqs. (72) and (73), together with the result $H_e(\Omega) = 0$ in the trapped region. Using the model collision operator defined in Eq. (62) we solve Eq. (73) to derive the parallel electron flow,

$$\left\langle \left\langle \frac{B u_{\parallel e}}{B} \right\rangle_{\theta} \right\rangle_{\Omega} = \frac{1}{2} \frac{\rho_{\theta e}}{L_n} v_{\text{the}} \left[\left(\frac{\omega}{\omega_{*e}} - (1 + \eta_e) \right) \left\langle \frac{\partial h}{\partial \chi} \right\rangle_{\Omega} - \frac{\omega}{\omega_{*e}} \right] \times (1 - \frac{3}{4} K \langle B^2 \rangle_{\theta}) + \frac{3}{16} K \langle B^2 \rangle_{\theta} \langle B \bar{u}_{\parallel e} \rangle_{\Omega} + \frac{3}{8} K \langle B^2 \rangle_{\theta} \langle B \bar{u}_{\parallel i} \rangle_{\Omega}, \quad (74)$$

where

$$\frac{\langle B \bar{u}_{\parallel e} \rangle_{\Omega}}{B} \left(1 - \frac{3}{8} K \langle B^2 \rangle_{\theta} \right) = \frac{\rho_{\theta e}}{L_n} v_{\text{the}} \left[\left(\frac{\omega}{\omega_{*e}} - \left(1 - \frac{\eta_e}{2} \right) \right) \left\langle \frac{\partial h}{\partial \chi} \right\rangle_{\Omega} - \frac{\omega}{\omega_{*e}} \right] \times \left(1 - \frac{3}{4} K \langle B^2 \rangle_{\theta} \right) + \frac{3}{4} K \langle B^2 \rangle_{\theta} \frac{\langle B u_{\parallel i} \rangle_{\Omega}}{B}. \quad (75)$$

(Since we shall not require the variation in the electron flow around a flux surface we have only evaluated the flux surface average here.)

Combining the flux surface average of the ion and electron flows, we obtain the total flux surface-averaged current perturbation:

$$\langle J_{bs} \rangle_{\Omega} = -nq_i \frac{\rho_{\theta e}}{L_n} v_{\text{the}} \left[1 + \frac{\eta_e}{4} + \tau^{-1} \left(1 - \frac{\eta_i}{2} \right) \right] \times \left(1 - \frac{3}{4} K \langle B^2 \rangle_{\theta} \right) \left\langle \frac{\partial h}{\partial \chi} \right\rangle_{\Omega}. \quad (76)$$

This is the bootstrap current perturbation caused by the presence of the magnetic island. However, this is not the total contribution to the current perturbation, and there is a part that varies around a flux surface and flux surface averages to zero, but which nevertheless contributes to the island width, Eq. (17). This is the ion polarization current and is calculated in the next section.

E. Current perturbation and saturated island width

In the previous section we have calculated the flux surface average of the current perturbation by directly calculating the particle distribution functions and constructing the $v_{\parallel}$ moment. In this section we calculate the part of the current perturbation that flux surface averages to zero, and that we identify with the ion polarization current. The most straightforward way to calculate this varying contribution is from the current continuity equation $\nabla \cdot \mathbf{J} = 0$, which can be constructed by integrating the drift-kinetic equations over veloc-

ity space, multiplying by the charge and summing over both species. Using quasineutrality simplifies this equation to give

$$\begin{aligned} \frac{1}{Rq} \frac{\partial J_{\parallel}}{\partial \theta} + k_{\parallel} \left. \frac{\partial J_{\parallel}}{\partial \xi} \right|_{\Omega} + \sum_j q_j \int d^3v \mathbf{v}_{dj} \cdot \nabla g_j \\ - \sum_j \frac{q_j^2}{m_j} \int d^3v \frac{(\mathbf{v}_d \cdot \nabla \Phi)}{v} \frac{\partial g_j}{\partial v} \\ = \sum_j \frac{q_j^2 \Phi}{T_j} \int d^3v F_{Mj} \mathbf{v}_d \cdot \nabla \Phi - \sum_j \frac{q_j}{n} \frac{dn}{d\chi} \frac{I}{Rq} \\ \times \left(1 - \frac{q_j \Phi}{T_j} \right) \int d^3v v_{\parallel} \frac{\partial}{\partial \theta} \left(\frac{v_{\parallel}}{\omega_{cj}} \right) \frac{\omega_{*j}^T}{\omega_{*j}} F_{Mj}. \end{aligned} \quad (77)$$

We are only interested in the leading-order contribution to the θ -averaged part of $J_{\parallel}$, which we write as $J_{\parallel}$. Therefore we multiply the above equation by Rq and integrate over a period in θ . The first term on the left and the terms on the right are then annihilated. The dominant contribution to $J_{\parallel}$ is then

$$\begin{aligned} Rq k_{\parallel} \left. \frac{\partial J_{\parallel}}{\partial \xi} \right|_{\Omega} = - \left\langle \sum_j q_j \int d^3v Rq \mathbf{v}_d \cdot \nabla g_j \right\rangle_{\theta} \\ \simeq -I \sum_j \frac{Rq}{\omega_{cj}} q_j \int d^3v v_{\parallel} \frac{\partial}{\partial \chi} \left\langle \frac{v_{\parallel}}{Rq} \frac{\partial g_j}{\partial \theta} \right\rangle_{\theta}, \end{aligned} \quad (78)$$

where terms of order ϵ^2 have been dropped. Clearly there is no contribution from $g_j^{(0,0)}$ and $g_j^{(0,1)}$ that is independent of θ . The contribution from $g_j^{(1,0)}$ θ averages to zero, so that the leading-order contribution comes from $g_j^{(1,1)}$ and must be evaluated using the $O(\Delta \delta_j)$ equation. In fact, the dominant contribution comes from the ions; for these the $O(\Delta \delta_i)$ equation yields

$$\frac{v_{\parallel}}{Rq} \frac{\partial g_i^{(1,1)}}{\partial \theta} = - \frac{\omega}{m \tilde{\psi}} \frac{dh}{d\Omega} Rq k_{\parallel} \left. \frac{\partial g_i^{(1,0)}}{\partial \xi} \right|_{\Omega} - k_{\parallel} v_{\parallel} \left. \frac{\partial g_i^{(1,0)}}{\partial \xi} \right|_{\Omega} + \dots, \quad (79)$$

where terms that do not contribute to Eq. (78) have not been shown. The first term on the right represents the time derivative and the $\mathbf{E} \times \mathbf{B}$ drift, while the second term represents the parallel streaming of ions along the perturbed field lines. A simple expression for the ion polarization current is obtained by considering the limit $\omega/k_{\parallel} v_{\text{thi}} > 1$, which is relevant when $w < \rho_{\theta i}$. This corresponds to the region of interest because in the opposite limit, $w > \rho_{\theta i}$, the polarization current is negligible compared to the bootstrap current and does not play a role in the island evolution. Thus we obtain

$$\begin{aligned} Rq k_{\parallel} \left. \frac{\partial J_{\parallel}}{\partial \xi} \right|_{\Omega} = -I \frac{Rq}{\omega_{ci}} q_i \frac{\omega}{m \tilde{\psi}} \\ \times \int d^3v v_{\parallel} \frac{\partial}{\partial \chi} \left(\frac{dh}{d\Omega} \left\langle Rq k_{\parallel} \left. \frac{\partial g_i^{(1,0)}}{\partial \xi} \right|_{\Omega} \right\rangle_{\theta} \right). \end{aligned} \quad (80)$$

Substituting the expression for $g_i^{(1,0)}$ from Eqs. (42) and (58) then yields the polarization current,

$$\begin{aligned} \left. \frac{\partial J_{\parallel}}{\partial \xi} \right|_{\Omega} = 24 \frac{q B^2 v_{\text{thi}}}{w_{\chi} w^3 \omega_{ci} s k_{\theta} L_n B_{\theta}^3} \frac{\omega(\omega - \omega_{*pi})}{\omega_{*i}} n q_i v_{\text{thi}} \\ \times \left[\frac{2}{3} \left\langle \frac{1}{\omega_{ci}} \right\rangle_{\theta} - \left\langle B \int_0^{\lambda_c} d\lambda \left\langle \frac{Rq}{\omega_{ci}} \right\rangle_{\theta} \right. \\ \left. \times \left\langle \frac{Rq}{\sqrt{1-\lambda B}} \right\rangle_{\theta}^{-1} \right] \frac{dh}{d\Omega} \frac{d^2 h}{d\Omega^2} \sin \xi, \end{aligned} \quad (81)$$

where $\omega_{*pi} = \omega_{*i}(1 + \eta_i)$. Employing a large aspect ratio expansion for the term in square brackets, we find

$$\int_0^{\lambda_c} B_0 d\lambda \left\langle \frac{1}{(1-\lambda B)^{1/2}} \right\rangle_{\theta}^{-1} = \frac{2}{3} + \left(I_p - \frac{1}{6} \right) (2\epsilon)^{3/2},$$

where

$$I_p = \frac{1}{2} \int_0^1 \left(\frac{2\pi}{K(k)} - 4 + k^2 \right) \frac{dk}{k^4},$$

and $K(k)$ is the complete elliptic integral of the first kind:

$$K(k) = \int_0^{\pi/2} \frac{d\alpha}{\sqrt{1 - k^2 \sin^2 \alpha}}.$$

Evaluating the integral numerically, we find $I_p = -0.219$. Clearly the square bracket in Eq. (81) is $O(\epsilon^{3/2})$, and on integrating, the following form for the variation of $J_{\parallel}$ around the flux surface is obtained:

$$\begin{aligned} \bar{J}_{\parallel} = -26\epsilon^{3/2} \left(\frac{\rho_{\theta i}}{w} \right)^3 \frac{w}{L_n} \frac{r}{s L_n} n q_i v_{\text{thi}} \\ \times \frac{1}{w_{\chi}^2} \frac{dh}{d\Omega} \frac{d^2 h}{d\Omega^2} \frac{\omega(\omega - \omega_{*pi})}{\omega_{*i}^2} [\cos \xi - \langle \cos \xi \rangle_{\Omega}] \\ + \langle J_{\text{bs}} \rangle_{\Omega}. \end{aligned} \quad (82)$$

We now have an expression for the full current perturbation:

$$\begin{aligned} \bar{J}_{\parallel} = 1.64\epsilon^{3/2} \frac{S}{Q^3} \left(\frac{\rho_{\theta i}}{w} \right)^3 \frac{w}{L_n} \frac{r}{s L_n} n q_i v_{\text{thi}} \\ \times \frac{\omega(\omega - \omega_{*pi})}{\omega_{*i}^2} [\cos \xi - \langle \cos \xi \rangle_{\Omega}] \\ + \frac{1.46}{QS} \epsilon^{1/2} \frac{\rho_{\theta i}}{L_n} n q_i v_{\text{thi}} \tau \left[1 + \frac{\eta_e}{4} + \tau^{-1} \left(1 - \frac{\eta_i}{2} \right) \right], \end{aligned} \quad (83)$$

where we have used the expression for $h(\Omega)$ from Eq. (55) together with $\langle J_{bs} \rangle_\Omega$ from Eq. (76) and defined the flux surface average,

$$S = \frac{1}{2\pi} \oint \frac{d\xi}{\sqrt{\Omega + \cos \xi}}. \quad (84)$$

Inserting this current perturbation into the dispersion relation, Eq. (17), yields the equation for the saturated island width,

$$\frac{\Delta'}{4} + 1.46\epsilon^{1/2}I_1 \frac{\beta_\theta}{w} \left(\frac{r}{sL_n} \right) \left[1 + \frac{\eta_e}{4} + \tau^{-1} \left(1 - \frac{\eta_i}{2} \right) \right] - 1.64\epsilon^{3/2}I_2 \left(\frac{\rho_{\theta i}}{w} \right)^2 \left(\frac{r}{sL_n} \right)^2 \frac{\beta_\theta}{w} \frac{\omega(\omega - \omega_{*pi})}{\omega_{*e}^2} = 0, \quad (85)$$

where $I_1 = 1.58$ and $I_2 = 1.42$ are numerical coefficients; we have also introduced the local poloidal β , $\beta_\theta = 8\pi n T_e / B_\theta^2$. This condition can be compared with the fluid result obtained in Ref. 14, which is

$$\frac{\Delta'}{4} + 1.46\epsilon^{1/2}G_1 \frac{r}{sL_n} \frac{\beta_\theta}{w} - G_2 \left(\frac{\rho_{\theta i}}{w} \right)^2 \left(\frac{r}{sL_n} \right)^2 \frac{\beta_\theta}{w} \frac{\omega(\omega - \omega_{*pi} - k\eta_i\omega_{*i})}{\omega_{*e}^2} = 0, \quad (86)$$

where k , G_1 , and G_2 are numerical coefficients. Neglecting temperature gradients, we find that the bootstrap current drive [i.e., the second term in Eq. (85)] is the same in both calculations and is independent of the collisionality regime. The differences occur in the third term, which is the contribution due to the ion polarization current. There are two: first, this term is $O(\epsilon^{3/2})$ smaller in the collisionless regime than the collisional regime; second, there is a difference in the frequency dependence as a result of neoclassical magnetic pumping, which leads to the term proportional to k in Eq. (86). This results from ion-ion collisions and is therefore not important in the collisionless regime. The effect does enter the collisionless calculation in the last term of the equation for the ion flow, Eq. (69), through the $H_i(\Omega)$ term. However, in contrast to the collisional theory, this is a flux surface-averaged term in the collisionless theory and therefore does not contribute to the polarization current.

Let us now return to Eq. (85), where we see the effect of the ion polarization current on the island evolution clearly depends on the propagation frequency. In particular, for $\omega < 0$ (i.e., in the electron direction) or $\omega > \omega_{*pi}$, the polarization current term is stabilizing and may provide a threshold island width, $w = w_c$. Thus, the determination of ω is crucial in deriving the criteria for the existence of neoclassical magnetic islands, and this is addressed in the following section.

F. Island propagation frequency

In the previous section we have outlined the importance of the island propagation frequency in determining the relevance of the ion polarization current for a threshold island width. This frequency is derived from Eq. (18), where we calculate $J_\parallel$ from the particle responses. Equation (18) is equivalent to toroidal torque balance, and the component of

$J_\parallel$ that is out of phase with the magnetic perturbation results from the dissipation in the plasma. In previous sections we have identified a narrow region of phase space close to the trapped/passing boundary, where collisional dissipation is important and modifies the distribution function. In the preceding analysis we were concerned with the component of current that is in phase with the magnetic perturbation, and the contribution to this from the “dissipation layer” is very small because only the barely passing particles contribute; hence, it was neglected. However, the dissipation layer provides the *only* source for the out-of-phase current and therefore must be calculated to determine the propagation frequency. This is the subject of the present section.

We first consider the passing electrons and seek the general solution to Eq. (71) in the dissipation layer, when all three terms must be treated as being of the same order. However, the narrowness of the layer implies the collision operator is dominated by the pitch-angle scattering term (which is proportional to $\partial^2 / \partial v_\parallel^2$). It is convenient to define a new pitch angle variable,

$$y = \frac{1 - \lambda B_0(1 + \epsilon)}{2\epsilon}, \quad (87)$$

which is such that $y = 0$ defines the trapped/passing boundary, with $y > 0$ corresponding to the passing region. In the narrow dissipation layer the collision operator is approximated by

$$C_e(g) \approx \frac{\nu_e}{2\epsilon^2} \frac{v_\parallel^2}{v^2} \frac{\partial^2 g}{\partial y^2}. \quad (88)$$

Equation (71) then reduces to

$$\frac{\sigma w_\chi}{|w_\chi|} \frac{(\alpha_e^p)^2}{2} \frac{\partial^2 \bar{h}_e^p}{\partial y^2} + \sqrt{\Omega + \cos \xi} \frac{\partial \bar{h}_e^p}{\partial \xi} \Big|_\Omega = \beta_e^p \sin \xi, \quad (89)$$

where

$$\frac{(\alpha_e^p)^2}{2} = \left| \frac{2\sqrt{2}}{\pi} \frac{\nu_e}{\epsilon \sqrt{2\epsilon k_\parallel v}} \right|, \quad (90)$$

$$\beta_e^p = -\frac{4\sqrt{\epsilon}}{\pi} \frac{\sigma v}{\omega_{\theta e}} \left[1 - \frac{1}{4s} \ln \left(4 \sqrt{\frac{\epsilon^{3/2} k_\parallel v_{the}}{\nu_e}} \right) \right] \times \frac{F_{Me}}{L_n} \frac{1}{w_\chi} \frac{dh}{d\Omega} \frac{(\omega - \omega_{*e}^T)}{\omega_{*e}}. \quad (91)$$

The superscript “ p ” indicates the passing phase space region and w_χ is defined to have the same sign as $(\chi - \chi_s)$. We have neglected weak logarithmic variations in y and substituted $\ln(4/\sqrt{y}) \rightarrow \ln(4\sqrt{\epsilon^{3/2} k_\parallel v_{the}/\nu_e})$. Note that β_e^p is odd in σ but even in $(\chi - \chi_s)$. Equation (89) must be solved subject to the boundary conditions that $\bar{h}_e^p$ matches onto the solutions that we have derived outside the dissipative layer. Thus, defining a stretched pitch angle variable, $z = \langle \sqrt{\Omega + \cos \xi} \rangle_\Omega^{1/2} y / \alpha_e^p$, and noting $H_e(\Omega) \approx 0$ in the layer, we have

$$\lim_{z \rightarrow \infty} \bar{h}_e^p = -2\beta_e^p (\sqrt{\Omega + \cos \xi} - \langle \sqrt{\Omega + \cos \xi} \rangle_\Omega). \quad (92)$$

It is convenient to write

$$f = \bar{h}_e^p + 2\beta_e(\sqrt{\Omega + \cos \xi} - \langle \sqrt{\Omega + \cos \xi} \rangle_\Omega), \quad (93)$$

and define a new variable x ,

$$x = \langle \sqrt{\Omega + \cos \xi} \rangle_\Omega \int_0^\xi \frac{d\xi'}{\sqrt{\Omega + \cos \xi'}}, \quad (94)$$

so that Eq. (89) becomes

$$\frac{1}{2} \frac{\partial^2 f}{\partial z^2} + \frac{\partial f}{\partial x} = 0, \quad (95)$$

with the boundary condition on f easily obtained from Eq. (92). Since $\bar{h}_e^p$, and therefore f , are periodic in ξ , they are also periodic in x , allowing a solution to be obtained as a Fourier expansion in x . Thus, we seek a solution of the form

$$f = \sum_k a_k(z) \cos kx + b_k(z) \sin kx. \quad (96)$$

Application of the boundary condition in Eq. (92) then yields

$$\begin{aligned} \bar{h}_e^p &= -2\beta_e^p(\sqrt{\Omega + \cos \xi} - \langle \sqrt{\Omega + \cos \xi} \rangle_\Omega) \\ &+ \sum_k \exp(-k^{1/2}z) [C_k^p \cos(k^{1/2}z + kx) \\ &+ D_k^p \sin(k^{1/2}z + kx)]. \end{aligned} \quad (97)$$

The sum over k is for integer $k > 0$ for $\sigma w_\chi > 0$ or $k < 0$ for $\sigma w_\chi < 0$ (for $k < 0$ the root $k^{1/2}$ is to be interpreted as $\sqrt{|k|}$). This expression can be substituted into the dispersion relation, Eq. (18). In particular, we find

$$\begin{aligned} \sum_{\pm} \oint \frac{\bar{J}_{\parallel e} \sin \xi}{\sqrt{\Omega + \cos \xi}} d\xi \\ = \sum_{\pm} \int d^3v q_e v_{\parallel} \oint \frac{\bar{h}_e^p \sin \xi}{\sqrt{\Omega + \cos \xi}} d\xi \\ = - \frac{\pi \epsilon q_e}{\langle \sqrt{\Omega + \cos \xi} \rangle_\Omega^{1/2}} \int_0^\infty v^2 dv \alpha_e^p \sum_{\pm} \sum_{k > 0} (\pm) \\ \times [C_k^p(\pm) - D_k^p(\pm) + C_{-k}^p(\pm) - D_{-k}^p(\pm)] I_k, \end{aligned} \quad (98)$$

where

$$I_k = \frac{1}{\sqrt{k}} \oint \frac{\sin \xi \sin kx}{\sqrt{\Omega + \cos \xi}} d\xi, \quad (99)$$

and $(\pm)$ indicates the coefficient calculated in the positive or negative $(\chi - \chi_s)$ region. To determine the current perturbation that is out of phase with the magnetic perturbation, it is necessary to calculate the C_k and D_k ; these are determined by matching to the trapped particle solution at the trapped/passing boundary. Thus, although the trapped particles themselves cannot carry a current, they do influence the current carried by the passing particles through this matching. Physically this represents the pitch-angle scattering of particles across the trapped/passing boundary, so that trapped particle information is carried to the passing particles.

We now solve the constraint equation for the trapped electrons, which is derived from the $O(\delta_e \Delta)$ equation by multiplying through by $Rq/|v_{\parallel}|$, summing over σ , and integrating between bounce points. The result is

$$\begin{aligned} \left\langle \frac{Rq}{|v_{\parallel}|} C(\bar{h}_e^t) \right\rangle_\theta - \frac{Rq}{m\tilde{\psi}} \frac{dh}{d\Omega} \omega \left\langle \frac{Rq}{|v_{\parallel}|} \right\rangle_\theta k_{\parallel} \left. \frac{\partial \bar{h}_e^t}{\partial \xi} \right|_\Omega \\ = \left(\langle |v_{\parallel}| \rangle_\theta + \omega_{ce} \frac{q_s}{q_s'} \frac{\partial}{\partial \chi} \left\langle \frac{|v_{\parallel}|}{\omega_{ce}} \right\rangle_\theta \right) \frac{4Rq}{w_\chi^2} \frac{I}{\omega_{ce}} \frac{F_{Me}}{n} \\ \times \frac{dn}{d\chi} \frac{(\omega - \omega_{*e})}{\omega_{*e}} \frac{dh}{d\Omega} k_{\parallel} \left. \frac{\partial \chi}{\partial \xi} \right|_\Omega. \end{aligned} \quad (100)$$

This constraint equation for the electrons is similar to that for the ions [Eq. (60)], though with a drive term proportional to $v_{\parallel}$, which is not important for the ions; however, this drive is crucial for the electrons for which it is large. The parallel motion averages to zero for the trapped electrons and it is therefore necessary to retain the propagation frequency; for the passing electrons, this was not important and was therefore dropped. For the low collision frequency that we consider, $\nu_e/\epsilon < \omega$, collisions are negligible over most of the trapped particle region. The trapped particle distribution function possesses a large discontinuity at the trapped/passing boundary, as shown schematically in Fig. 3(b). This is resolved by the collisional term, which becomes important in a layer close to the trapped/passing boundary, but in the trapped region, of width $(\nu_e/\omega)^{1/2}$. Furthermore, because of the large discontinuity resulting from the trapped particle constraint, the pitch-angle scattering is enhanced at the trapped/passing boundary, so that matching here results in a larger current carried by the passing particles than their small $(\nu_e/k_{\parallel} v_{\parallel e})^{1/2}$ boundary layer width would suggest. Thus trapped electron effects are crucial in calculating the out-of-phase current, and it is necessary to solve for them in the trapped “layer” region. In this region, close to the trapped/passing boundary, the constraint equation simplifies:

$$\frac{(\alpha_e^t)^2}{2} \frac{\partial^2 \bar{h}_e^t}{\partial y^2} + \sqrt{\Omega + \cos \xi} \left. \frac{\partial \bar{h}_e^t}{\partial \xi} \right|_\Omega = \beta_e^t \sin \xi, \quad (101)$$

where

$$\frac{(\alpha_e^t)^2}{2} = \frac{\sqrt{2}}{4} \left(\frac{w_\chi}{h'} \right) \left| \frac{v_e}{\epsilon \omega} \right| \left[\ln \left(4 \sqrt{\frac{\epsilon \omega}{\nu_e}} \right) \right]^{-1}, \quad (102)$$

$$\begin{aligned} \beta_e^t &= - \frac{\epsilon}{\sqrt{2}} \left[\ln \left(4 \sqrt{\frac{\epsilon \omega}{\nu_e}} \right) \right]^{-1} \left[1 - \frac{1}{4s} \ln \left(4 \sqrt{\frac{\epsilon \omega}{\nu_e}} \right) \right] \\ &\times \frac{\bar{k}_{\parallel} v^2}{\omega \omega_{\theta e}} \frac{F_{Me}}{L_n} \frac{(\omega - \omega_{*e}^T)}{\omega_{*e}}, \end{aligned} \quad (103)$$

and $\bar{k}_{\parallel}$ is defined to have the same sign as $(\chi - \chi_s)$ so that β_e^t is odd in $(\chi - \chi_s)$. Note that we have selected one particular sign of ω here; both signs give the same result and there is no loss of generality. We have again neglected weak logarithmic variations in y and substituted $\ln(4/\sqrt{y}) \rightarrow \ln(4\sqrt{\epsilon \omega/\nu_e})$. We must solve Eq. (101), subject to the constraint that the

solution matches onto the collisionless solution for deeply trapped particles. Thus, following the method of solution for the passing particles we have

$$\begin{aligned} \bar{h}_e^t &= -2\beta_e^t(\sqrt{\Omega + \cos \xi} - \langle \sqrt{\Omega + \cos \xi} \rangle_\Omega) \\ &+ \sum_{k>0} \exp(k^{1/2}z) [C_k^t \cos(k^{1/2}z - kx) \\ &+ D_k^t \sin(k^{1/2}z - kx)], \end{aligned} \quad (104)$$

where for trapped particles we have

$$z = \frac{\langle \sqrt{\Omega + \cos \xi} \rangle_\Omega^{1/2}}{\alpha_e^t} y, \quad (105)$$

so that $z \rightarrow -\infty$ corresponds to the deeply trapped region.

The problem has now been reduced to a determination of the coefficients C_k and D_k by matching at the trapped/passing boundary, $z=0$. Three matching conditions must be imposed there:

$$\sum_\sigma \sigma \bar{h}_e^p = 0, \quad (106)$$

$$\sum_\sigma \bar{h}_e^p = 2\bar{h}_e^t, \quad (107)$$

$$\sum_\sigma \frac{\partial \bar{h}_e^p}{\partial y} = 2 \frac{\partial \bar{h}_e^t}{\partial y}. \quad (108)$$

The first two conditions simply result from matching $\bar{h}_e$ across the boundary and making use of the fact that $\bar{h}_e^t$ must be independent of σ . The third condition states that trapped particles must be scattered into passing orbits (of either sign of σ) at the same rate as passing particles are scattered into trapped orbits. The application of these boundary conditions involves some straightforward, but tedious, algebra. For each matching condition the $\cos kx$ and $\sin kx$ components are projected out to yield the following set of six coupled equations for the C_k and D_k :

$$\begin{aligned} C_k^p - C_{-k}^p &= r^2 \left[1 - \frac{1}{4s} \ln \left(4 \sqrt{\left| \frac{\epsilon^{3/2} k_{\parallel} v_{\text{the}}}{\nu_e} \right|} \right) \right] \\ &\times \left[1 - \frac{1}{4s} \ln \left(4 \sqrt{\left| \frac{\epsilon \omega}{\nu_e} \right|} \right) \right]^{-1} \beta_e^t M_k, \\ D_k^p + D_{-k}^p &= 0, \\ C_k^p + C_{-k}^p - 2C_k^t &= -\beta_e^t M_k, \\ D_k^p &= -D_k^t, \\ C_k^p + C_{-k}^p &= -2r(C_k^t + D_k^t), \\ C_k^p - C_{-k}^p + 2D_k^p &= -2r(C_k^t - D_k^t), \end{aligned} \quad (109)$$

where β_e^p has been expressed in terms of β_e^t through a small quantity r^2 :

$$r^2 = \frac{4\sqrt{2}}{\pi} \frac{1}{\sqrt{\epsilon}} \left(\frac{1}{w_\chi} \frac{dh}{d\Omega} \right) \left| \frac{\omega}{k_{\parallel} v} \right| \ln \left(4 \sqrt{\left| \frac{\epsilon \omega}{\nu_e} \right|} \right), \quad (110)$$

and M_k is the flux surface average,

$$M_k = \frac{8}{\pi} \langle \sqrt{\Omega + \cos \xi} \rangle_\Omega \int_0^\pi \cos kx \, d\xi. \quad (111)$$

The same equations hold for $\chi > \chi_s$ or $\chi < \chi_s$, though β_e^t switches sign. Expanding for small r we arrive at the required results,

$$C_k^p + C_{-k}^p = -r\beta_e^t M_k, \quad (112)$$

$$D_k^p + D_{-k}^p = 0. \quad (113)$$

Substituting these back into the current equation, Eq. (98), we find that the D_k terms do not contribute, while the C_k terms give rise to

$$\begin{aligned} \sum_\pm \int_{-1}^\infty d\Omega \oint \frac{\bar{J}_{\parallel e} \sin \xi}{\sqrt{\Omega + \cos \xi}} d\xi \\ = -G_s \epsilon^{3/2} n q_e v_{\text{the}} \frac{\rho_{\theta e}}{L_n} \sqrt{\left| \frac{\nu_e}{\epsilon \omega} \right|} \left[\ln \left(4 \sqrt{\left| \frac{\epsilon \omega}{\nu_e} \right|} \right) \right]^{-1/2} \\ \times \left[1 - \frac{1}{4s} \ln \left(4 \sqrt{\left| \frac{\epsilon \omega}{\nu_e} \right|} \right) \right] \frac{[\omega - \omega_{*e}(1 + \eta_e/4)]}{\omega_{*e}}, \end{aligned} \quad (114)$$

where

$$\begin{aligned} G_s &= \frac{8}{\pi^{5/2}} \Gamma\left(\frac{7}{4}\right) \int_1^\infty \frac{d\Omega}{\sqrt{QS}} \sum_{k>0} \frac{1}{\sqrt{k}} \oint \cos kx \, d\xi \\ &\times \oint \frac{\sin \xi \sin kx}{\sqrt{\Omega + \cos \xi}} d\xi = 4.66. \end{aligned} \quad (115)$$

The flux surface averages Q and S are defined in Eqs. (54) and (84), respectively.

Equation (114) provides the contribution of the electrons to the dispersion relation. The ions are much simpler, and in fact we shall see that they yield a negligible contribution. The trapped ions do not influence the out-of-phase current carried by the passing particles in the layer region. To see this we consider the passing particle layer region for the ions where the distribution function satisfies

$$\frac{\alpha_i^2}{2} \frac{\partial \bar{h}_i^p}{\partial y^2} + \sqrt{\Omega + \cos \xi} \frac{\partial \bar{h}_i^p}{\partial \xi} \Big|_\Omega = \beta_i \sin \xi, \quad (116)$$

with

$$\frac{\alpha_i^2}{2} = \frac{1}{2\sqrt{2}} \left| \frac{\nu_i}{\epsilon \omega} \right| \frac{w_\chi}{h'} \left[\ln \left(4 \sqrt{\left| \frac{\epsilon \omega}{\nu_i} \right|} \right) \right]^{-1}, \quad (117)$$

$$\begin{aligned} \beta_i &= -\frac{\pi}{\sqrt{2}} \left(\frac{h'}{w_\chi} \right) \frac{F_{\text{Mi}}}{L_n} \left[\ln \left(4 \sqrt{\left| \frac{\epsilon \omega}{\nu_i} \right|} \right) \right]^{-1} \\ &\times \sqrt{2\epsilon} \frac{\sigma v}{\omega_{\theta i}} \frac{(\omega - \omega_{*i}^T)}{\omega_{*i}}. \end{aligned} \quad (118)$$

Again, we have chosen a particular sign for ω , but the final answer is independent of this choice. The solution for the ion equation in the passing region is

$$\begin{aligned} \bar{h}_i^p = & -2\beta_i(\sqrt{\Omega + \cos \xi} - \langle \sqrt{\Omega + \cos \xi} \rangle_\Omega) \\ & + \sum_{k>0} \exp(-k^{1/2}z) [C_k^p \cos(k^{1/2}z + kx) \\ & + D_k^p \sin(k^{1/2}z + kx)], \end{aligned} \quad (119)$$

which satisfies the boundary condition that it matches to the collisionless distribution function in the deeply passing region ($z \rightarrow \infty$). Here z has been defined as in Eq. (105), but with $\alpha_e^t \rightarrow \alpha_i$. For the ions the collisional term in the layer equation, Eq. (116), does not depend on σ , and this leads to a simpler solution and the following expression for the out-of-phase current:

$$\begin{aligned} \oint \frac{J_{\parallel i} \sin \xi}{\sqrt{\Omega + \cos \xi}} d\xi = & -\frac{\pi q_i \epsilon}{\langle \sqrt{\Omega + \cos \xi} \rangle_\Omega^{1/2}} \int_0^\infty v^3 dv \alpha_i \\ & \times \sum_\sigma \sum_{k>0} I_k \sum_\sigma \sigma (C_k^p - D_k^p). \end{aligned} \quad (120)$$

This can be evaluated by considering the passing particles only and applying the boundary condition that the passing particle solution must satisfy $\sum_\sigma \sigma h_i^p = 0$ at $y=0$. The following conditions are derived for the C_k^p and D_k^p :

$$\sum_\sigma \sigma C_k^p = -M_k \sigma \beta_i, \quad (121)$$

$$\sum_\sigma \sigma D_k^p = 0. \quad (122)$$

These expressions can be substituted directly into Eq. (120) and there is no need to calculate the trapped ion response. The following expression for the ion contribution to the out-of-phase current is found:

$$\begin{aligned} \sum_\pm \int_{-1}^\infty d\Omega \oint \frac{\bar{J}_{\parallel i} \sin \xi}{\sqrt{\Omega + \cos \xi}} \\ = -\frac{\pi^2}{4} G_s \epsilon^{3/2} n q_i v_{thi} \frac{\rho_{\theta i}}{L_n} \sqrt{\frac{\nu_i}{\epsilon \omega}} \\ \times \left[\ln \left(4 \sqrt{\frac{\epsilon \omega}{\nu_i}} \right) \right]^{-3/2} \frac{[\omega - \omega_{*i}(1 + \eta_i/4)]}{\omega_{*i}}. \end{aligned} \quad (123)$$

The total out-of-phase current is obtained by summing the contributions from the ions and electrons. However, the electron contribution dominates by a factor $\sim \sqrt{\nu_e/\nu_i}$ so we ignore the ion contribution. Substitution into the dispersion relation, Eq. (18), then yields the following expression for the propagation frequency of the islands:

$$\omega = \omega_{*e}(1 + \eta_e/4). \quad (124)$$

It should be noted that we calculated the ion response in the dissipation layer, assuming $w < \epsilon^{-1/2} \rho_{\theta i}$, so that $\omega > k_{\parallel} v_{\parallel i}$. For larger island widths, the calculation of the ion response in the layer follows that of the electrons described above. The ion contribution to the out-of-phase current will be

$O[(\nu_i/\epsilon \omega)^{1/2}]$ and therefore negligible, even for larger islands, so that Eq. (124) is valid for all island widths $w > \epsilon^{1/2} \rho_{\theta i}$.

Substituting this frequency into Eq. (85) for the island width, we find that the polarization current is stabilizing and therefore gives rise to a threshold value for saturated islands, as discussed in the following section.

III. CONCLUSION

We have developed a self-consistent, neoclassical kinetic theory for the existence of magnetic islands in tokamak plasmas in the low collisionality regime. Two particular mechanisms are found to dominate the evolution of magnetic islands when their width exceeds the ion banana width. First, the bootstrap current drive for the island exists in the “collisionless” theory [$\nu_j < \max(\omega, k_{\parallel} v_{\parallel j})$], as well as the collisional fluid theory [$\nu_j > \max(\omega, k_{\parallel} v_{\parallel j})$], and has the potential to sustain magnetic islands of large saturated width. Second, the ion polarization current also affects the island evolution, though because of its dependence on the island propagation frequency its effect is rather subtle. There are two main differences in the polarization currents that arise in the two collision frequency regimes. In the fluid case the ion polarization current is enhanced over the FLR contribution by a large factor, $\sim q^2/\epsilon^2$, while in the low collision frequency regime the enhancement is much smaller, $\sim q^2/\sqrt{\epsilon}$. Second, in the fluid regime the dissipation is expected to be dominated by the “magnetic pumping” resulting from ion–ion collisions so that the islands might be expected to propagate in the ion diamagnetic direction.¹⁴ The polarization drift can then be destabilizing, in which case there is no threshold island width for island growth results. However, in the low collision frequency regime the barely passing electrons within a region of pitch angle space $\sim \sqrt{\nu_e/\epsilon^{3/2} k_{\parallel} v_{the}}$ dominate the dissipation and the islands propagate in the electron diamagnetic direction, provided external torques can be neglected. The ion polarization drift is then stabilizing and a threshold island width exists. These points are illustrated by the equation for the saturated island width, which we have derived in Eq. (85). From this equation we see that there are two possible scenarios, depending on the sign of the polarization current term. These are illustrated in Fig. 4, where the left-hand side of Eq. (85), $f(w)$, is plotted as a function of w when the polarization current term is positive [Fig. 4(a)] and negative [Fig. 4(b)]. In Fig. 4(a) it can be seen that when the polarization current term is positive, only one solution for the saturated island width exists. By retaining the time evolution of the island width this solution can be shown to be stable (see Refs. 2 and 3, for example). Thus, if $w = w_{sat}$ is the stable solution, then an island with $w < w_{sat}$ will grow with time, while an island with $w > w_{sat}$ will shrink. This evolution is indicated by the arrows on the curves. Clearly, in this situation an initial small magnetic perturbation will be magnified and result in an island with a large saturated width, $w = w_{sat}$. No threshold mechanism is therefore provided by the polarization current in this case. We now turn to Fig. 4(b), which is the analogous diagram to Fig. 4(a), but with the polarization current term assumed to be negative. In

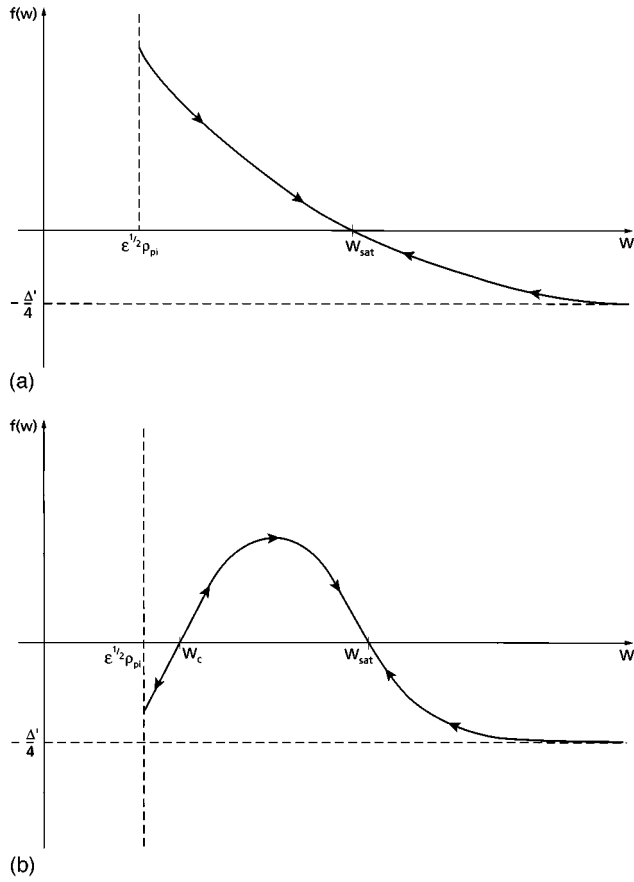


FIG. 4. Sketch of the dispersion relation, Eq. (85), in the form $f(w)=0$ in the cases when (a) the ion polarization current is destabilizing and (b) when it is stabilizing.

this case two positive solutions for w exist. The upper solution, $w=w_{\text{sat}}$, corresponds to the stable solution identified in Fig. 4(a). Thus, we deduce that the lower solution $w=w_c$ is unstable, the arrows indicating the direction of island evolution. Clearly an initial magnetic perturbation with an island width $w>w_c$ will evolve toward the saturated solution, w_{sat} , while islands of width $w<w_c$ will decay away. Thus $w=w_c$ provides a threshold island width below which magnetic islands are not expected to be seen. In the collisionless case, ω is in the electron diamagnetic direction so that the polarization current term is negative and the situation corresponds to that in Fig. 4(b). For the higher, stable solution, $w=w_{\text{sat}}$, the polarization current, which falls off as w^{-3} , is small so that an expression for w_{sat} results from the balance between the bootstrap current drive and the Δ' damping. However, for the lower, unstable solution $w=w_c$, where w_c is comparable to the banana width, the polarization drift has much more influence and the Δ' term is negligible. Thus, a threshold island width, w_c , above which saturated islands can exist is predicted; w_c results from a balance of the bootstrap current drive and the polarization current damping:

$$w_c = \epsilon^{1/2} \left(\frac{r}{sL_n} \right)^{1/2} \left(\frac{(1 + \eta_e/4)[\tau(1 + \eta_e/4) + 1 + \eta_i]}{1 + \eta_e/4 + \tau^{-1}(1 - \eta_i/2)} \right)^{1/2} \rho_{\theta i}. \quad (125)$$

The validity of our result for the threshold is at the limit

of the applicability of our expansion procedure, which considers islands of width $w \gg \epsilon^{1/2} \rho_{\theta i}$, the ion banana width. The threshold that we have found is in fact of the order of the banana width. However, a more rigorous calculation that treats the region $w \sim \epsilon^{1/2} \rho_{\theta i}$ accurately would involve much more complicated physics. First, finite ion Larmor radius effects become important so that a gyrokinetic model must be used to describe the ion response. Second, the higher-order, $O(\epsilon^2)$, corrections to the parallel flows will be required, and this will involve a more careful treatment of the flux surface averages of equilibrium variables. Furthermore, at lower island widths it may be possible for the passing electrons to have a Landau resonance so that the dissipation may not be governed by the collisional process discussed here. Finally, finite banana width effects will need to be considered. It is not clear that such a treatment will be analytically tractable, and a numerical model, based on the ideas developed in this work, may be necessary. However, we expect that the result that we have derived for the threshold magnetic island width is qualitatively correct. This is because the FLR ion polarization drift has the same form as the neoclassically enhanced version derived here, and this form would be valid when the island width exceeds the ion Larmor radius. It should provide an additional but small damping to that calculated here. On the other hand, it is important to realize that extra drives and dissipation mechanisms (e.g., Landau damping) may be operative for islands with a width comparable to, or smaller than, the ion Larmor radius,²⁴ and our theory does not rule out stable solutions for these “micro-islands.”

As well as the impact of bootstrap-driven magnetic islands on confinement, further interest stems from the evolution of the so-called “locked” modes. These occur where a relatively low-amplitude mode is generated because of “error fields” due to imperfections in tokamak coils or that are deliberately induced to observe their effect. This low-amplitude mode rotates in the laboratory frame with some characteristic frequency. However, under certain circumstances the mode can “lock” (i.e., stop rotating) at which point it is seen to grow to a very large saturated state. The threshold mechanism discussed here, which depends crucially on the island rotation frequency, provides a potential model for interpreting the dynamics of these locked modes.

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